

# The Effect of Purchase Intention Interruption on Novelty-Seeking: Evidence from a Large Coffee Chain\*

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## Abstract

Temporary supply disruptions can do more than postpone demand: they may change what consumers choose after access is restored. We study this effect using 18 temporary exogenous store closures at a large digital coffee retailer. Because the same closure is more likely to interrupt an intended purchase for some consumers than for others, we use pre-closure behavior to predict each consumer's purchase intention during the closure and estimate a triple-differences design comparing high- and low-predicted-intention consumers. Our main finding concerns product exploration. We find that novelty-seeking, defined as the share of distinct products purchased that the consumer has not purchased before, increases by 3.13 percentage points for low-intention consumers but decreases by 1.02 percentage points for high-intention consumers. The resulting high-minus-low difference is  $-4.15$  percentage points, equivalent to 8.3% of the pre-period mean. An alternative measure based on recently introduced products yields a similar  $-4.17$ -percentage-point reduction. The decline in exploration is also stronger among consumers with lower baseline novelty-seeking. These patterns are consistent with an interrupted purchase goal increasing the salience of familiar choices. More broadly, our findings show that recovery from a temporary disruption has a demand-composition dimension: consumers may return without returning to their previous level of product exploration. Firms should therefore monitor product discovery and new-product adoption alongside sales recovery after disruptions.

**Keywords:** Purchase Interruption, Product Exploration, Variety Seeking, Retail Disruptions, Goal-Directed Consumption

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# 1 Introduction

Temporary disruptions are common in retail and service markets. Stores close, products stock out, capacity becomes unavailable, and consumers are temporarily prevented from completing purchases they otherwise might have made. A natural view of such disruptions is that they primarily shift demand across time or across sellers: purchases are delayed, redirected, or lost, and consumers resume their previous behavior once access returns. We study a different possibility. When a temporary constraint blocks an intended purchase, the disruption may change not only *when* consumers purchase, but also *what* they choose after the constraint is removed.

A large marketing literature shows that supply-side disruptions can have consequences beyond the transaction that is immediately lost. Store closures affect chain sales, store switching, customer retention, and activity across offline and online channels (Haans and Gijsbrechts, 2010; Seenivasan, Sudhir, and Talukdar, 2016; Ye and Shankar, 2025). Related research on stockouts and service-capacity constraints shows that temporary unavailability can affect both current and subsequent purchasing behavior and that these responses vary across customers and products (Anderson, Fitzsimons, and Simester, 2006; Che, Chen, and Chen, 2012; Jing and Lewis, 2011; von Wangenheim and Bayón, 2007). Together, this literature establishes that temporary restrictions on access can have persistent demand consequences. Our question differs in two respects. First, we distinguish consumers for whom the same supply disruption is likely to *bind* against an intended purchase from consumers who are exposed to the disruption but are unlikely to purchase during it. Second, rather than focusing primarily on whether, where, or how much consumers subsequently purchase, we study whether a blocked purchase changes the composition of subsequent demand through consumers’ propensity to explore products.

This distinction also connects our setting to research on state dependence and dynamic consumer choice. Prior work shows that current product choices can depend on past choices through inertia, reinforcement, or variety seeking, and that purchase timing and product choice may evolve jointly (Chintagunta, 1999; Lattin, 1987; Simonson, 1990). More recent work uses plausibly exogenous disruptions to choice opportunities to separate state dependence from persistent consumer heterogeneity. For example, Levine and Seiler (2023) use hurricane-induced stockouts to study whether forced brand switching changes subsequent brand choice, while Levine and Hristakeva (2025) use a temporary grocery-store disruption and differences in displaced shopping trips to study persistence in store choice. We extend this line of inquiry by shifting attention

from a *realized past choice* to an *intended but unrealized choice*. Our setting asks whether preventing consumers from executing a likely purchase can itself create a state that affects how broadly they choose once purchasing becomes possible again.

Our focus on product exploration further relates to research showing that variety seeking depends on both consumers' prior choices and the environment in which choices are made. Assortment composition affects assortment perceptions, store choice, purchasing, and customer retention (Boatwright and Nunes, 2001; Borle, Boatwright, Kadane, Nunes, and Shmueli, 2005; Briesch, Chintagunta, and Fox, 2009; Broniarczyk, Hoyer, and McAlister, 1998). Shopping context can also alter the breadth of consumer choice: Chintala, Liaukonytė, and Yang (2024), for example, document lower basket variety and greater overlap across online grocery trips, while Bronnenberg, Klein, and Xu (2024) show that consumers' time constraints shape grocery-shopping behavior. This literature emphasizes that product exploration is not a fixed consumer trait. We contribute by studying whether a temporary interruption to the purchasing process changes subsequent exploration even after consumers regain access to the retailer.

We examine these questions using panel data from a large digital coffee retailer covering 779,744 consumers and 260 stores from June 1, 2020 through December 31, 2021. Our empirical variation comes from 18 temporary store closures during the COVID-19 pandemic. These closures sharply restrict access to particular stores for a limited period and then end, allowing us to observe the same consumers before the disruption and after the store reopens.

The central empirical challenge is that exposure to a closure is not equivalent to having a purchase interrupted. A consumer who was unlikely to purchase during the closure experiences the same store-level disruption as a consumer who was highly likely to purchase, but the constraint binds only for the latter. We therefore use consumers' pre-disruption purchase histories to predict their probability of purchasing during each closure window. We then estimate a triple-differences design comparing (i) treated and control consumers, (ii) behavior before and after reopening, and (iii) consumers with high and low predicted purchase intention. Under the identifying assumptions described below, the high-minus-low treatment-effect differential isolates the incremental response associated with interrupting a likely purchase rather than the general effect of exposure to a store closure.

Our main result concerns the composition of demand after reopening. We measure novelty-seeking as the share of distinct products purchased within a purchasing window that the consumer has not purchased previously. Among low-predicted-intention consumers, the treated-control post-reopening effect is

an increase of 3.13 percentage points in novelty-seeking. Among high-predicted-intention consumers, the corresponding point estimate is a decline of 1.02 percentage points. Their difference is  $-4.15$  percentage points, equivalent to 8.3% of the 49.78

The purchase-frequency results tell a different story. Purchase frequency increases by 0.67 percentage points per calendar day for high-predicted-intention consumers and by 0.09 percentage points for low-predicted-intention consumers, producing a high-minus-low difference of 0.58 percentage points per day. However, the corresponding pretrend test rejects equality. We therefore treat this result as descriptive evidence that purchasing activity recovers after reopening rather than as a clean causal estimate. This distinction is economically important: transaction activity can return while the composition of consumers' purchases remains altered.

Several additional results reinforce the product-choice finding. First, replacing consumer-specific novelty with new-product-seeking—the share of purchased products that have appeared recently in the transaction data—produces a DDD of  $-4.17$  percentage points. The finding therefore does not depend solely on defining novelty relative to each consumer's own purchase history. Second, the negative pattern also appears when the outcome is restricted to non-coffee products, although smaller samples and diagnostic failures prevent us from giving those estimates the same causal interpretation as the main result. Third, the interruption-related reduction in novelty-seeking is concentrated among consumers who were less exploratory before the closure. Holding baseline purchasing history fixed, the fitted DDD becomes substantially less negative as baseline novelty-seeking rises. This heterogeneity suggests that a blocked purchase does not create a uniform movement toward familiar products; instead, it reinforces familiar choice most strongly among consumers for whom familiar products were already relatively important.

We interpret these patterns through the lens of goal persistence. When a likely purchase cannot be completed, the underlying purchase goal may remain active. Once access returns, completing that interrupted goal may make the familiar purchase path relatively salient, reducing consumers' willingness to deviate toward unfamiliar products. This interpretation is consistent with research showing that unfulfilled goals remain cognitively active and that interruptions can affect subsequent decision processes (Dhar, Huber, and Khan, 2007; Fitzsimons, 2000; Gollwitzer, 1990; Kruglanski, Shah, Fishbach, Friedman, Chun, and Sleeth-Keppler, 2002; Masicampo and Baumeister, 2011). We do not claim that the empirical design separately identifies this psychological mechanism. Rather, goal persistence provides one account of the combination of reduced exploration, stronger effects among consumers with lower baseline novelty-seeking, and

continued purchasing after reopening.

Our findings contribute to the marketing literature in three ways. First, we add a new dimension to research on temporary supply disruptions. Existing studies demonstrate that closures, stockouts, and capacity constraints affect sales, retention, channel use, store choice, and future transactions (Anderson et al., 2006; Haans and Gijsbrechts, 2010; Jing and Lewis, 2011; Seenivasan et al., 2016; von Wangenheim and Bayón, 2007; Ye and Shankar, 2025). We show that the consequences of a disruption also depend on whether the constraint is likely to bind against an existing purchase intention. This distinction between *exposure* and *interruption* reveals heterogeneity that average closure effects can mask.

Second, we extend research on state dependence by showing that the relevant state need not originate from a completed past choice. A purchase that a consumer likely intended to make but was prevented from making can also predict subsequent choice behavior. In this sense, temporary constraints can leave a behavioral footprint even after the supply restriction itself has disappeared. Our findings therefore complement work using exogenous switching or trip displacement to study persistence in realized store and brand choices (Levine and Hristakeva, 2025; Levine and Seiler, 2023).

Third, we contribute to research on variety seeking and product discovery by identifying a disruption-induced change in exploration. Prior research has emphasized how purchase histories, timing, assortments, and shopping environments shape variety and repeat choice (Boatwright and Nunes, 2001; Borle et al., 2005; Broniarczyk et al., 1998; Chintagunta, 1999; Chintala et al., 2024; Simonson, 1990). We show that a temporary interruption can narrow exploration after consumers return, even though the interruption itself has ended. This distinction has a direct managerial implication. Recovering traffic or transaction frequency does not necessarily mean that customer demand has fully recovered: product discovery and new-product adoption may remain depressed among consumers whose intended purchases were interrupted. Firms managing temporary closures, stockouts, service interruptions, or other access constraints should therefore monitor the *composition* of returning demand in addition to its level.

## **2 Conceptual Motivation and Empirical Predictions**

### **2.1 Interrupted Goals and Familiar-Choice Salience**

A temporary supply constraint can affect consumers in two conceptually different ways. All consumers associated with a closed store are exposed to the disruption, but the constraint is more likely to bind for

consumers who would otherwise have purchased during the closure. For these consumers, the closure does not simply remove access to the store; it prevents the completion of a likely purchase. We propose that this distinction between exposure and interruption can affect consumers' subsequent product choices after access is restored.

Our behavioral interpretation draws on research on unfinished goals and counterfactual thinking. Unfulfilled goals can remain cognitively accessible after goal pursuit is interrupted (Masicampo and Baumeister, 2011), while counterfactual thinking—the mental representation of how a past event could have unfolded differently—can influence subsequent intentions and behavior (Epstude and Roese, 2008). Related consumer research shows that responses to product unavailability depend on consumers' commitment to the intended option (Fitzsimons, 2000). Research on implemental mindsets further suggests that goal pursuit can focus attention on completing an intended action rather than reconsidering alternatives (Dhar et al., 2007).

Applied to our setting, a consumer whose likely purchase is prevented by a store closure may mentally represent the purchase that would have occurred had the store remained open. Because routine purchases are grounded in prior consumption experience, this representation may keep the consumer's familiar purchase path relatively salient. The interrupted goal need not correspond to one specific product. It may instead be broader—for example, completing one's usual consumption activity—so that products already associated with that routine become relatively accessible when the consumer returns.

We refer to this response as *familiar-choice salience*. Counterfactual thinking and goal persistence provide one possible cognitive foundation for it, but our empirical design does not directly observe either process and therefore does not separately identify the psychological mechanism. The empirical implication we study is narrower: if interrupting a likely purchase makes the familiar purchase path relatively salient, consumers should become less likely to explore unfamiliar products after reopening.

This account does not generate an unambiguous prediction for purchase frequency. An active purchase goal may encourage consumers to return after reopening, but substitution during the closure, discouragement, or other responses to the disruption may offset this effect. The prediction is instead about the *composition* of demand conditional on purchasing: consumers whose likely purchases are interrupted should shift relatively toward familiar products and therefore exhibit lower novelty-seeking.

## 2.2 A Parsimonious Framework

We formalize this intuition with a reduced-form framework that maps directly to the empirical design. Let  $i$  index consumers and  $t$  relative periods around the closure associated with consumer  $i$ . Because each consumer appears in only one closure event, we suppress a separate event index. Let  $\text{Post}_t$  indicate periods after reopening,  $\text{Treated}_i$  indicate whether the consumer is exposed to a focal store closure, and let  $p_i \in [0, 1]$  denote the ex-ante probability that consumer  $i$  would have made a purchase during the closure window absent the closure. A larger  $p_i$  therefore represents a greater likelihood that the closure interrupts a purchase rather than merely exposing the consumer to a store-level disruption.

Let  $Y_{it}^N$  denote novelty-seeking conditional on making a purchase, and let  $\vartheta_i$  summarize the consumer's underlying tendency to explore unfamiliar products. We represent expected novelty-seeking as

$$\mathbb{E}[Y_{it}^N \mid \text{Purchase}_{it} = 1] = \mu_{it}(\vartheta_i) - \lambda(\vartheta_i) (\text{Post}_t \times \text{Treated}_i \times p_i), \quad (1)$$

where  $\mu_{it}(\vartheta_i)$  captures baseline novelty preferences and changes in product choice unrelated to the interruption-specific familiar-choice channel. The function  $\lambda(\vartheta_i)$  captures the additional salience of familiar choices generated when a likely purchase is interrupted.

The behavioral account imposes two restrictions:

$$\lambda(\vartheta_i) > 0, \quad \lambda'(\vartheta_i) < 0. \quad (2)$$

The first restriction captures the central idea: a likely purchase interruption shifts subsequent choice toward familiar products. The second captures heterogeneity in this response. Consumers with a stronger underlying preference for novelty have a weaker familiar-choice tendency, so the interruption-induced shift toward familiar products is smaller. In the empirical analysis, we use consumers' baseline novelty-seeking to proxy for this underlying heterogeneity.

To connect the framework to our empirical comparison, let  $\Delta_T$  denote the treated-minus-control difference and  $\Delta_P$  denote the post-minus-pre difference. We require that, absent the interruption-specific familiar-choice channel, the resulting difference-in-differences does not vary systematically with predicted purchase intention:

$$\frac{\partial}{\partial p_i} \Delta_T \Delta_P \mu_{it}(\vartheta_i) = 0. \quad (3)$$

This restriction captures the identifying logic of the triple-differences design: absent the interruption channel, general post-reopening differences between treated and control consumers should not systematically vary with ex-ante purchase intention.

Under equations (1)–(3), the treated-control post response becomes more negative as purchase intention rises:

$$D_p(\vartheta_i) \equiv \frac{\partial}{\partial p_i} \Delta_T \Delta_P \mathbb{E} [Y_{it}^N \mid \text{Purchase}_{it} = 1] = -\lambda(\vartheta_i) < 0. \quad (4)$$

**Prediction 1** (Purchase interruption reduces novelty-seeking). *Compared with consumers with lower purchase intention during the closure, consumers with higher purchase intention should exhibit a lower treated-control post-reopening response in novelty-seeking. Thus, the high-minus-low triple-differences effect on novelty-seeking should be negative.*

The framework is written using the continuous purchase probability  $p_i$  because this makes the intensity of likely interruption explicit. Our main empirical specification instead classifies consumers into high- and low-predicted-intention groups. For any  $p_H > p_L$ , the corresponding finite-difference implication is

$$D(p_H, p_L; \vartheta_i) = -\lambda(\vartheta_i)(p_H - p_L) < 0. \quad (5)$$

Thus, the binary high-minus-low DDD used in the main analysis is a discrete counterpart of the continuous prediction. We also estimate the continuous-score specification as a robustness check.

The second restriction in equation (2) generates a heterogeneity prediction. Differentiating the interruption effect with respect to novelty preference gives

$$\frac{\partial D_p(\vartheta_i)}{\partial \vartheta_i} = -\lambda'(\vartheta_i) > 0. \quad (6)$$

Because  $D_p(\vartheta_i) < 0$ , this inequality implies that the interruption effect becomes less negative as the consumer's underlying preference for novelty increases.

**Prediction 2** (Baseline novelty attenuates the interruption effect). *The negative interruption-related effect on novelty-seeking should be stronger among consumers with lower baseline novelty-seeking and weaker among consumers who are already more exploratory.*

The framework is intentionally limited to product choice conditional on purchasing. It does not impose



a signed prediction on purchase frequency, nor does it imply that counterfactual thinking or goal persistence is uniquely responsible for the empirical patterns. Its purpose is to formalize the behavioral implication that motivates our tests: when a temporary constraint is more likely to interrupt an intended purchase, subsequent choice shifts toward the consumer’s familiar purchase path, with the strongest shift among consumers whose prior behavior is already less exploratory.

### 3 Setting and Data

We use administrative purchase records from a large Chinese quick-service coffee chain whose customers order primarily through a mobile application. The data cover 19 calendar months from June 1, 2020 to December 31, 2021 and contains 7,312,498 orders, represented by 10,631,250 product-line records, from 779,744 customers at 260 stores in a major city. We observe the customer, purchase date, store and product for each order.

Temporary store closures due to Covid-19 pandemic lockdowns create the treatment variation. We identify store closures of stores with no transactions in a consecutive days varying from 10 to 30 days, leading to 18 identified closures. The detailed applied sample screens are summarized in Appendix [A.1](#). The temporary closures happened mostly in July and August, 2021, because of a large round of lockdowns in the city.

We next introduce how we construct treatment and control groups. A store closure can potentially affect all consumers in our record, as consumers can purchase in any store. To exclude infrequent consumers, we first restrict our attention to consumers with at least 5 purchases before the closures happened, and this constitutes around 30% of our sample. We assign treatment status to consumers who always purchase at a focal store and that store experienced a closure. Specifically, the focal store of a consumer is defined as the one where the consumer made at least 80% of purchases among all her purchases. For each store closure identified, we select 5 similar stores (e.g. have a similar setup date) that operate normally through all time. We treat the consumers whose focal stores are among these selected control stores as control customers. These restrictions make the focal store a meaningful source of treatment exposure.

The resulting sample contains 18 closures and 40,148 customers in total, including 7,390 treated customers and 32,758 controls customers. For each temporary closure, we consider four pre-closure and four post-reopening windows, with time window length the same as the closure length.

### 3.1 Classifying Predicted Purchase intention

The central challenge in this setting is that not all treated customers are equally affected by a store closure. While all customers assigned to a treated store are exposed to the disruption, only a subset would have made a purchase during the closure window absent the interruption. We separate closure exposure from likely purchase interruption by classifying members according to predicted purchase intention. A likely interrupted purchase occurs only when a high-predicted-intention consumer is also treated.

We classify the intention of each member with a predictive XGBoost model trained on pre-closure behavior. The model estimates whether the member would make at least one purchase in a pre-period window, based on his history before this window. We train one model for each closure duration, since the probability of at least one purchase mechanically depends on the number of days in the prediction window. Training uses relative pre-periods  $-4$  through  $-2$ , validation uses the holdout pre-period  $-1$ , and an additional audit evaluates control customers in period 0, when their matched stores remain open. For treated customers, the corresponding period-0 target absent the focal-store closure is counterfactual, so we use the predicted probability. We classify a member as high predicted purchase intention when this probability is at least 0.5.

The features use only information available before the prediction window: recent purchase frequency, recency since the last purchase, total pre-period purchase days, store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Table 1: Purchase-intention classifier performance on the final cohort

| Evaluation sample        | $N$    | Prevalence | ROC-AUC | PR-AUC | Brier | Calibration slope |
|--------------------------|--------|------------|---------|--------|-------|-------------------|
| Control, period 0        | 32,758 | .263       | .781    | .583   | .172  | .471              |
| Control, pre-period $-1$ | 32,758 | .308       | .814    | .677   | .164  | .536              |
| Treated, pre-period $-1$ | 7,390  | .314       | .818    | .697   | .162  | .537              |

Notes: All rows use the exact final 18-event cohort. Prevalence is the observed share purchasing at least once in the evaluation window. ROC-AUC measures ranking performance; PR-AUC should be compared with prevalence, the performance of an uninformative ranking. The Brier score is the mean squared probability error, for which lower values are better. The corresponding constant-prevalence Brier scores are .194, .213 and .216. Perfect calibration has intercept zero and slope one; the estimated intercepts are  $-.599$ ,  $-.309$  and  $-.276$ , respectively. Period-0 outcomes are not observable for treated customers absent the closure, so that validation is conducted only among controls.

Table 1 reports validation results on the exact final 18-event cohort. The period-0 control audit is the closest observable counterpart to the treated counterfactual: its ROC-AUC is 0.781 and its PR-AUC is 0.583, compared with an observed purchase rate of 0.263. Its Brier score is 0.172, below the 0.194 score

from assigning every observation the sample purchase rate. The model therefore ranks likely purchasers meaningfully. We use the binary classification in our main econometric specification, and retain the continuous score as a robustness check. Appendix A.4 reports confusion matrices, calibration, duration-specific performance, predictive tradeoffs across cutoffs and score overlap.

The prediction of purchase intention is central to our empirical design. The comparison between high- and low-predicted-intention customers separates general closure exposure from the additional response associated with a likely purchase interruption, subject to the identifying assumptions below. We focus on how temporary store closures affect consumer behavior with a focus on consumers' propensity to explore new products.

### 3.2 Data Statistics

The final panel contains 321,184 potential observations in the form of closure-consumer-period, among which we have 99,644 observations of purchases. The trained classifier places 2,041 treated customers, or 27.6%, above the high-predicted-intention threshold. Appendix A.2 reports the complete observation flow.

Table 2: Four-cell descriptive statistics before closure

| Variable                                        | Low predicted intention |                | High predicted intention |                |
|-------------------------------------------------|-------------------------|----------------|--------------------------|----------------|
|                                                 | Control                 | Treated        | Control                  | Treated        |
| Customer-closure customers                      | 23,934                  | 5,349          | 8,824                    | 2,041          |
| Predicted purchase-intention probability        | .135<br>(.141)          | .132<br>(.139) | .773<br>(.157)           | .785<br>(.158) |
| Pre-period purchase frequency                   | .022<br>(.035)          | .021<br>(.033) | .127<br>(.116)           | .114<br>(.112) |
| Pre-period novelty-seeking                      | .604<br>(.392)          | .602<br>(.392) | .450<br>(.309)           | .498<br>(.305) |
| customers with at least one pre-period purchase | 12,317                  | 2,745          | 8,798                    | 2,034          |
| Share of pre-period windows with a purchase     | .214                    | .211           | .751                     | .753           |
| Closure duration in days                        | 16.2<br>(5.6)           | 16.9<br>(6.5)  | 18.8<br>(6.4)            | 21.7<br>(7.2)  |

Notes: The unit is a customer-closure consumer. Standard deviations are in parentheses. Pre-period purchase frequency is first averaged over relative periods  $-4$  through  $-1$  within a consumer. Pre-period novelty-seeking is averaged over purchasing windows and is reported for customers with at least one pre-period purchase. The purchase share is the fraction of the four pre-period windows in which the customer purchases at least one product. High and low groups differ mechanically in purchase propensity; the relevant descriptive comparison is treated versus control within each predicted-intention group. Closure duration is repeated at the consumer level and is shown descriptively, without customer-level balance inference.

Table 2 displays the four columns that form the DDD. High-intention customers make at least one purchase in about three quarters of their pre-period windows, compared with about one fifth for low-intention

customers. Because novelty-seeking is defined only in purchasing windows, these purchase probabilities also determine entry into the product-choice sample. Within the low-intention stratum, treated and control customers have similar pre-period purchase frequency and novelty-seeking. Differences are larger in the high-intention stratum, particularly for closure duration and pre-period novelty, so the empirical design relies on fixed effects and the group-specific pretrend tests rather than level balance alone.

Figure 1 presents the raw closure shock in purchase frequency, where relative period 0 is the closure window and y-axis indicates the number of daily purchases. Daily purchases of the treated consumers falls from 0.039 in period  $-1$  to 0.007 during the closure, while the control mean is 0.036 during the same relative period. In the first post-reopening window, the treated and control means are 0.039 and 0.039, respectively. The raw paths therefore verify that the treated consumers were really affected by the closure and that purchasing returns near the comparison path after reopening.

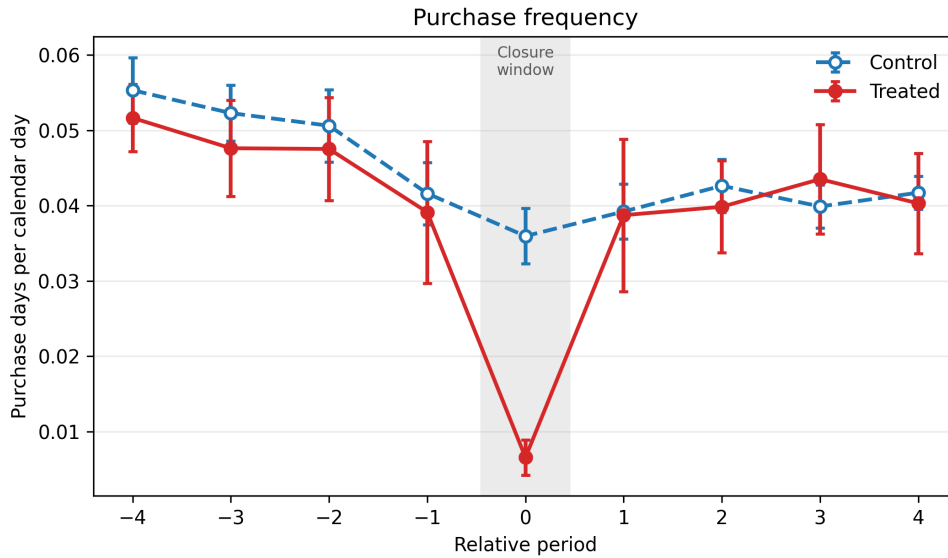


Figure 1: Store closures sharply reduce treated purchase frequency during the closure window

Notes: The figure plots customer-weighted pooled means for the 18-closure analysis sample; each relative period contains 32,758 control and 7,390 treated customer-closure customers. Bars are 95% confidence intervals based on standard errors clustered by closure event. Relative period 0 is the closure window. Purchase frequency is the number of days with any purchase anywhere in the observed chain divided by the number of calendar days in the event-length window. Treatment is assigned through the focal pre-closure store, but the outcome includes purchases at other chain stores.

## 4 Empirical Strategy and Results

### 4.1 Outcomes

Let  $i$  index consumers,  $e$  closure events,  $t$  relative periods and let  $L_e$  the duration in days of closure event  $e$ . For each event, we construct relative periods  $t = -4, \dots, 4$ , where each period is a window of length  $L_e$  and period 0 is the closure window. The main regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This construction produces a member-event-period panel in which each member-event contributes up to eight observations to the purchase-frequency analysis. We define two main outcomes on these windows.

**Purchase frequency.** We measure purchase intensity using purchase frequency, defined as the number of days with at least one purchase at any chain store per calendar day within each event-length window. For consumer  $i$  and closure event  $e$ , we define purchase frequency as mean daily purchases during the window

$$Y_{iet}^P = \frac{\#Purchases_{it}}{L_e}, \quad (7)$$

Normalizing by closure length makes purchase frequency comparable across closure events of different durations.

**Novelty-seeking.** To measure exploration relative to a consumer's own purchase history, let  $\mathcal{J}_{iet}$  denote the set of distinct products that consumer  $i$  purchases in event window  $t$ , and let  $\mathcal{H}_{iet}^-$  denote the products the consumer purchased before that window. We define novelty-seeking as

$$Y_{iet}^N = \frac{|\mathcal{J}_{iet} \setminus \mathcal{H}_{iet}^-|}{|\mathcal{J}_{iet}|}. \quad (8)$$

The numerator is the unique products purchased in the current window that are not purchased before, while the denominator is the unique products purchased in the current window. A higher novelty-seeking therefore indicates that a larger share of the purchased distinct products new to that consumer. Novelty-seeking is defined only when a consumer purchases at least one product in the window; otherwise the denominator is zero and the outcome is missing. It therefore describes product composition conditional on purchasing, whereas purchase frequency captures the extensive margin. Appendix D.2 uses new-product-seeking as an alterna-

tive novelty-seeking measurement on consumers' preference towards products that are newly introduced by the company.

Having defined the outcomes and predicted purchase-intention status, we next plot the raw variation behind the triple-differences design. Figure 2 shows customer-weighted mean novelty-seeking and closure-clustered confidence intervals for the four estimation subgroups, omitting the closure window. In the low-predicted-intention panel, treated and control means are similar before closure, but the gap enlarges in the post-reopening windows, indicating a positive treatment effect. In the high-predicted-intention panel, the gap between treated and control means begins at a higher level but merely evolves in the pre-periods. The gap merely changes after reopening, so it's already clear that the treatment effect is much smaller than the one in low-predicted group.

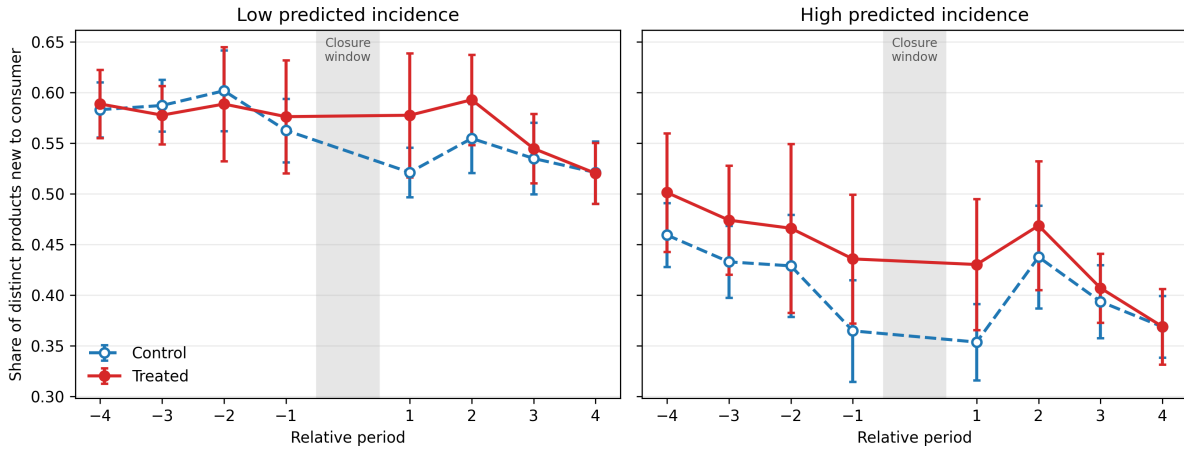


Figure 2: Raw novelty-seeking paths motivate the high-minus-low comparison

Notes: The figure plots customer-weighted pooled means for the 18-closure analysis sample. Bars are 95% confidence intervals based on standard errors clustered by closure event. Novelty-seeking is the share of distinct products purchased in a window that are new to the consumer and is observed only in purchasing windows. Across the eight plotted periods, observed-window counts range from 4,227 to 6,354 for control-low, 919 to 1,397 for treated-low, 4,918 to 7,469 for control-high and 1,171 to 1,714 for treated-high customers. Relative period 0, the closure window, is omitted to match the main regressions.

Novelty-seeking is observed only in windows with a purchase, so sample entry varies across cells and relative periods. Figure 3 provides the analogous four-cell plot for the extensive margin: the probability of at least one purchase in the window. Within each predicted-intention stratum, the treated and control paths are close, although purchase probability falls after reopening, particularly among high-predicted-intention customers. The figure makes changing sample entry transparent, but similar treated and control means do not establish that the observed purchasers have the same composition. The novelty estimates therefore retain a conditional-on-purchase interpretation.

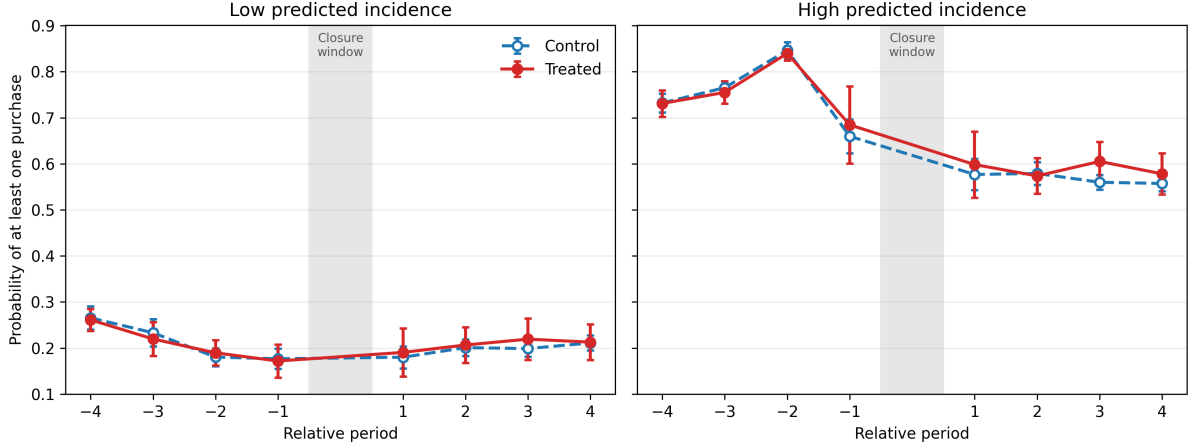


Figure 3: Purchase probability evolves similarly for treated and control customers within predicted-intention groups

Notes: The figure plots the share of customer-closure windows containing at least one purchase anywhere in the observed chain. This is also the share of windows in which novelty-seeking is defined. Each relative period contains 23,934 control-low, 5,349 treated-low, 8,824 control-high and 2,041 treated-high customers. Bars are 95% confidence intervals based on standard errors clustered by closure event. Means are customer weighted. Relative period 0 is omitted to match the main regressions.

## 4.2 Triple Difference-in-Differences

To estimate the incremental response associated with interrupting a likely purchase, we compare (i) treated versus control customers, (ii) behavior before and after reopening, and (iii) high versus low predicted purchase intention.

The logic follows the canonical triple-differences design in [Gruber \(1994\)](#), who studies group-specific mandates by comparing affected and unaffected groups across states with and without policy changes before and after adoption. In this setting, the third difference removes shocks common to adopting states, shocks common to the affected group, and aggregate time effects, leaving the component specific to the treated affected group. In our setting, closure exposure is the treated-unit dimension, predicted purchase intention is the affected-group dimension, and the post-reopening period is the time dimension. Recent applications use the same idea in richer interaction settings. [Gehring and Schaudt \(2026\)](#), for example, interpret the interaction between drought exposure and insurance rollout as a triple-difference-style estimator with continuous treatment dimensions and emphasize that the main threat is an omitted factor that moves with the interaction term. More closely, [Levine and Hristakeva \(2025\)](#) study the impact of temporary store closure on consumers who likely planned a visit during the closure, focusing on visiting frequency later. This perspective motivates the coefficient decomposition and pretrend tests below.

Formally, we estimate the following pooled specification:

$$Y_{iet} = \delta^B \cdot \text{Post} \times \text{Treated} + \beta \cdot \text{Post} \times \text{High} + \delta^D \cdot \text{Post} \times \text{Treated} \times \text{High} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}, \quad (9)$$

where  $Y_{iet}$  denotes the outcome of interest for member  $i$  in period  $t$  of event  $e$ . The indicator *Post* captures periods after reopening, *Treated* identifies customers exposed to a store closure, and *High* identifies customers with high predicted purchase intention during the closure window. This indicator is defined ex ante for both treated and control member-events; only treated high-predicted-intention customers face a likely interrupted purchase. The specification includes member-closure fixed effects  $\phi_{ie}$ , relative-period fixed effects  $\omega_t$  and calendar-month fixed effects  $\gamma_m$ . Because treatment exposure is generated by a common store-closure shock, standard errors are clustered by closure event.

We estimate the model pooling 18 closure events instead of estimating closure-specific coefficients. Closure windows can overlap in calendar time; calendar-month fixed effects absorb common monthly conditions, but do not make events equally weighted. The pooled coefficient is implicitly weighted by the customer-period observations contributed by each event, which range with the number of eligible customers. The low-predicted-intention treatment effect is  $\delta^B$ . The high-predicted-intention treatment effect is  $\delta^B + \delta^D$ . Thus  $\delta^D$  is the high-minus-low treatment-effect differential, not the total effect for the high group. The parameter  $\beta$  is a high-minus-low post shift common to treated and control customers; it is not a treatment effect.

#### 4.2.1 Identification Assumptions and Diagnostic Tests

The causal interpretation of the high-minus-low differential requires two central assumptions.

First, we require *comparable general closure effects*. A closure may affect customers for reasons other than interrupting a likely purchase, for example by changing convenience, promotions, substitution opportunities or perceptions of the store. The low-predicted-intention comparison captures these general effects. The high-predicted-intention comparison contains the same general effects plus the consequence of a likely interrupted purchase. If general closure effects would otherwise be comparable across the two groups, subtracting the low-group response from the high-group response isolates the incremental response associated with the interruption. However, we admit that this assumption is untestable.

Second, *parallel triple trends* must hold: absent the closure, the treated-control gap for high-predicted-



intention customers would have evolved like the corresponding gap for low-predicted-intention customers. Table 2 compares treated and control customers within each predicted-intention group. The low-intention cells have similar pre-period outcomes, while the high-intention cells differ more in pre-period novelty and closure duration. These level differences motivate the fixed effects and group-specific pretrend checks below.

We use the pre-closure periods to test whether the comparison underlying the DDD was already changing before the focal store closed. The event study uses the last pre-closure window, period  $-1$ , as the reference. For each earlier window including periods  $-4$ ,  $-3$  and  $-2$ , we estimate whether the treated-control gap changed more for high- than for low-predicted-intention customers, relative to period  $-1$ . We then test whether these three pre-closure estimates are jointly equal to zero. For purchase frequency, they are jointly distinguishable from zero ( $p = 0.035$ ). For novelty-seeking, the three pre-closure estimates range from 1.60 to 2.64 percentage points, but every 95% confidence interval includes zero and the joint test does not reject that all three estimates equal zero ( $p = 0.841$ ). Thus, we do not detect a differential pretrend in novelty-seeking. This result supports the parallel-triple-trends assumption on novelty-seeking. Appendix B reports the full estimates.

### 4.3 Main Results

We decompose the main estimates in Table 3: Panel A reports the treated-control post contrast separately for low- and high-predicted-intention customers, and Panel B reports their difference. This layout keeps the total high-group effect distinct from the DDD. Appendix Tables W13 report the full set of estimates.

**Purchase frequency.** The low-predicted-intention treated-control post contrast is 0.09 percentage points per calendar day (SE 0.24;  $p = 0.711$ ). The high-group contrast is 0.67 points (SE 0.45;  $p = 0.152$ ), or 13.6% of the pre-period mean. Their difference is 0.58 points per day (SE 0.32;  $p = 0.086$ ), about 11.8% of the pre-period mean. The point estimates describe a relative post-reopening rebound, but none of the group-specific contrasts is statistically distinguishable from zero at the 5% level under closure-clustered inference. Together with the rejected high-minus-low purchase pretrend, this evidence is descriptive rather than a clean estimate of an interruption-induced recovery.

Table 3: Decomposition of the main triple-differences results

| Effect                                                           | Purchase frequency                  | Novelty-seeking                       |
|------------------------------------------------------------------|-------------------------------------|---------------------------------------|
| <i>Panel A. Closure effects within predicted-intention group</i> |                                     |                                       |
| Low predicted purchase intention ( $\delta^B$ )                  | .0009<br>(.0024)<br>[.7107]         | .0313*<br>(.0178)<br>[.0976]          |
| High predicted purchase intention ( $\delta^B + \delta^D$ )      | .0067<br>(.0045)<br>[.1524]         | -.0102<br>(.0072)<br>[.1745]          |
| <i>Panel B. Incremental interruption contrast</i>                |                                     |                                       |
| High minus low ( $\delta^D$ )                                    | <b>.0058*</b><br>(.0032)<br>[.0860] | <b>-.0415**</b><br>(.0184)<br>[.0380] |
| Pre-period outcome mean                                          | .0493                               | .4978                                 |
| Observations                                                     | 321,184                             | 99,644                                |
| $R^2$                                                            | .562                                | .416                                  |
| Within $R^2$                                                     | .013                                | .001                                  |

Notes: Panel A reports treated-control post contrasts within low- and high-predicted-intention customers. Panel B reports their difference, the triple-differences coefficient; boldface marks this focal estimand. Purchase frequency is purchase days per calendar day. Novelty-seeking is the share of distinct products purchased in a window that the consumer had not purchased before that window and is defined only for windows with a purchase. The high-group effects are estimated in algebraically equivalent parameterizations and have their own cluster-robust standard errors. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . All regressions include member-closure, relative-period and calendar-month fixed effects and omit the closure window.  $R^2$  includes the absorbed fixed effects; within  $R^2$  is calculated after absorbing them.

**Novelty-seeking.** For low-predicted-intention customers, the treated-control contrast is a 3.13-percentage-point increase in novelty-seeking (SE 1.78;  $p = 0.098$ ) and is statistically distinguishable from zero at the 10% level. One natural explanation is the history-dependent definition of novelty: as consumers accumulate purchases, fewer products remain untried, causing novelty-seeking to decline over time, as shown by the pre-period pattern. During the closure, control consumers continue to purchase and expand their familiar-product sets, whereas treated consumers have fewer opportunities to do so. Consequently, after reopening, a larger share of the products purchased by treated consumers may still be classified as novel. Relatedly, control consumers may have already tried products introduced during the closure, while those products remain untried—and therefore novel—for treated consumers. These forces can generate a positive treated-control contrast for the low group and represent general consequences of store closure rather than the effect of interrupting a likely purchase. In contrast, the high-group effect is a 1.02-point decline (SE 0.72;  $p = 0.175$ ), although it is not statistically distinguishable from zero. The focal causal estimate is the difference between these two contrasts: the DDD is  $-4.15$  points (SE 1.84;  $p = 0.038$ ), or 8.3% of the pre-period outcome

mean. Thus, the incremental effect associated with interrupting a likely purchase more than offsets the 3.13-point increase observed among low-predicted-intention customers, leaving the high-group point estimate slightly negative. Under parallel triple trends and comparable general closure effects, this result indicates that interrupting a likely purchase causally reduces novelty-seeking relative to the product-choice response that would otherwise follow a closure. Economically, consumers whose likely purchases are interrupted concentrate more heavily on familiar products after reopening, rather than exhibiting the increased exploration observed among consumers whose purchases were unlikely to have been interrupted.

## 5 Evidence on Behavioral Mechanism and Alternative Explanations

Our main results on novelty seeking are consistent with prediction 1 based on counterfactual thinking and goal persistence. However, they do not directly prove that our proposed mechanism is the underlying cause of the causal triple-interaction estimate. In this section, we further build confidence in our mechanism by conducting a heterogeneity analysis to test prediction 2. We also evaluate two alternative explanations for the causal triple-interaction estimate. The evidence weighs against the notification channel and against differential return timing as a quantitatively important explanation, subject to the measurement limitations discussed below.

### 5.1 Heterogeneity by Baseline Novelty-Seeking

Consumers with lower baseline novelty-seeking reveal a stronger orientation toward familiar products and established routines. When an expected purchase is interrupted, the salient counterfactual may therefore be more focused on the familiar product that would otherwise have been chosen. If our mechanism holds, the interruption should reduce novelty-seeking more strongly among these consumers.

**Baseline novelty-seeking and distribution.** For each customer, we first calculate novelty-seeking separately in relative periods  $-8$  through  $-1$  and then take the mean across periods in which the customer purchased. Each relative period has the length of its closure event. We define the consumer’s *baseline orders*,  $O_{ie}$ , as the total number of distinct orders that customer  $i$  placed during these same eight windows. We require  $O_{ie} \geq 5$  to reduce the influence of very short purchase histories. The raw novelty measure ranges from zero, when every purchased product in these windows is familiar, to one, when every distinct product

purchased in each purchasing window is new relative to the customer's earlier history.

Figure 4 shows the resulting distribution across 18,525 eligible customers. The mean is 0.508, the standard deviation is 0.269 and the median is 0.500. The 10th and 90th percentiles are 0.167 and 0.889; 3.3% of customers are at zero and 6.8% are at one. Thus, the analysis uses broad variation over the full support rather than a comparison driven by a narrow part of the distribution. Among the 17,866 customers with an observed novelty outcome, every treatment-by-predicted-intention cell contains all 18 closures; cell sizes range from 1,489 to 7,951 customers, and cell means range from 0.453 to 0.574.

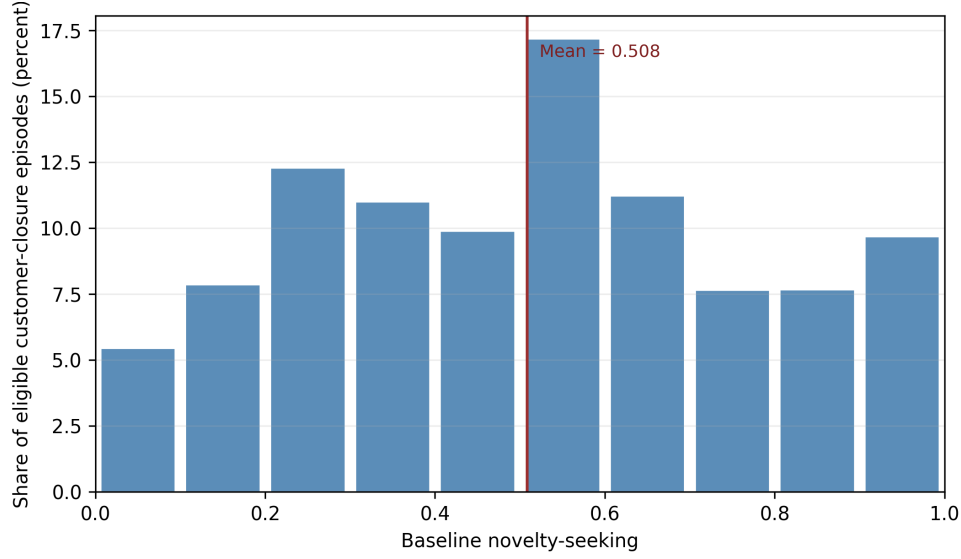


Figure 4: Distribution of baseline novelty-seeking

Notes: The figure shows ten equal-width bins of the raw consumer-level baseline novelty measure. The measure is the mean of within-period novelty-seeking over purchasing periods from relative period  $-8$  through  $-1$ . Eligible customers have at least five orders across these eight closure-length periods. The last bin includes one. The vertical line marks the sample mean. The sample contains 18,525 eligible customer-closure customers.

By definition, novelty-seeking is naturally conditional on at least one purchase made. Moreover, novelty-seeking decreases as more orders are made because there are less unique products remaining to be tried. The raw data also show the correlation. Across the 18,525 eligible customers, the correlation between baseline novelty and  $\log(1 + O_{ie})$  is  $-0.329$ . Mean baseline novelty is 0.620 among customers with exactly five orders and 0.365 among customers with at least 20 orders; the share at the upper boundary of one falls from 19.3% to 0.1% across these groups. This model-free association may reflect both genuine differences between frequent and infrequent customers and the greater sampling variability of novelty measured from few orders. Either source can make order history independently relevant for heterogeneity in the post-period response. As a result, we include baseline orders into our following econometric specification.

**Econometric specification.** Let  $N_{ie}$  denote standardized baseline novelty-seeking. We standardize the baseline-order count as  $Q_{ie} = [\log(1 + O_{ie}) - \overline{\log(1 + O)}] / \text{SD}[\log(1 + O)]$ , where  $O$  is the mean baseline orders. The logarithm compresses the right tail of the order-count distribution, and standardization gives  $Q_{ie}$  mean zero and unit standard deviation. Retaining the Post, Treated and High definitions from equation (9), our specification is

$$\begin{aligned}
Y_{iet}^N = & \delta^B(\text{Post} \times \text{Treated}) + \beta(\text{Post} \times \text{High}) + \delta^D(\text{Post} \times \text{Treated} \times \text{High}) \\
& + \eta_0(\text{Post} \times N_{ie}) + \eta_T(\text{Post} \times \text{Treated} \times N_{ie}) \\
& + \eta_H(\text{Post} \times \text{High} \times N_{ie}) + \theta^D(\text{Post} \times \text{Treated} \times \text{High} \times N_{ie}) \\
& + \text{Post} \times Q_{ie} (\kappa_0 + \kappa_T \text{Treated} + \kappa_H \text{High} + \kappa_D \text{Treated} \times \text{High}) \\
& + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet},
\end{aligned} \tag{10}$$

where  $Y_{iet}^N$  is novelty-seeking conditional on purchase. The first line is exactly the main DDD specification for this outcome.

The second and third line add the complete Post-by-baseline-novelty interaction hierarchy.  $\eta_0$  is used to control for the change in novelty seeking that is purely caused by heterogeneity.  $\eta_T$  measures the incremental effect of a higher baseline novelty-seeking on the treatment effect of the low-intention group.  $\eta_H$  measures the incremental effect of a higher baseline novelty-seeking on the triple interaction term and is our main coefficient of interest.

The line involving  $Q_{ie}$  allows post-period changes to vary flexibly with the amount of baseline purchase history. In the control, low-predicted-intention cell,  $\kappa_0$  is the change in the post-period outcome gradient associated with a one-standard-deviation increase in log baseline orders. The coefficients  $\kappa_T$  and  $\kappa_H$  allow that gradient to differ for treated and high-predicted-intention customers, respectively, while  $\kappa_D$  allows the high-minus-low DDD itself to vary with baseline orders. The levels of  $Q_{ie}$  and its group interactions are time-invariant and therefore absorbed by  $\phi_{ie}$ ; their interactions with Post are the estimable terms. Last, member-closure fixed effects  $\phi_{ie}$  absorb all time-invariant consumer characteristics,  $\omega_t$  are relative-period fixed effects and  $\gamma_m$  are calendar-month fixed effects. This complete hierarchy prevents  $\theta^D$  from capturing heterogeneity associated with how many orders were available to measure baseline novelty. Because  $N_{ie}$  and  $Q_{ie}$  are standardized,  $\delta^D$  is the high-minus-low DDD at mean baseline novelty and mean baseline order

history. The parameter  $\theta^D$  is the change in this DDD for a one-standard-deviation increase in baseline novelty, holding baseline order history fixed. We also estimate an otherwise identical specification after deleting the entire line involving  $Q_{ie}$ . Both specifications use the same sample, outcome, fixed effects and closure-event clustering.

**Results.** Table 4 reports the adjusted and unadjusted estimates side by side. With the baseline-order hierarchy, a one-standard-deviation increase in baseline novelty raises the DDD by 2.94 percentage points (SE = 1.47; one-sided  $p = 0.031$ ). Without that hierarchy, the gradient remains positive at 2.29 points (SE = 1.43; one-sided  $p = 0.064$ ). The direction and approximate magnitude therefore do not depend on the baseline-order adjustment, although the unadjusted estimate does not meet the 5% one-sided threshold. We prefer the adjusted specification because the raw association above shows that measured baseline novelty varies sharply with the number of orders used to construct it.

One standard deviation is 0.269 on the raw baseline-novelty scale. In the adjusted specification, the DDD at mean baseline novelty and order history is  $-1.30$  points (SE = 2.00). Holding order history at its mean, the linear fit moves from a  $-4.24$ -point DDD at raw baseline novelty of about 0.240-one standard deviation below the mean, to a  $+1.63$ -point DDD at raw baseline novelty of about 0.777-one standard deviation above the mean. This two-standard-deviation movement changes the fitted interruption response by 5.88 points. These fitted contrasts summarize the adjusted slope; they are not separately estimated subgroup effects.

The adjusted and unadjusted event-study interactions do not reject a differential pretrend ( $p = 0.561$  and  $p = 0.409$ , respectively), supporting a causal interpretation of  $\theta^D$ , with details in Appendix Figure W4. The positive coefficients match the prediction 2 based on our proposed mechanism: lower-baseline-novelty customers exhibit the more negative interruption-related response.

## 5.2 Alternative Explanations

Besides our proposed mechanism on counterfactual thinking and goal persistence, other alternative explanations could also explain a negative novelty DDD. First, high- and low-intention customers may be under different intensities of exposure to new-product notifications in the post-reopening periods, which leads to difference in treatment effect on novelty seeking between the two groups. Second, after store reopening, high- and low-intention customers may face different assortments when they return, and thus a negative novelty DDD is caused by a supply-side constraint. The analyses below do not support either ex-

Table 4: Baseline novelty-seeking and the interruption-related novelty response

|                                                                  | With $Q_{ie}$ adjustment<br>(1) | Without $Q_{ie}$ adjustment<br>(2) |
|------------------------------------------------------------------|---------------------------------|------------------------------------|
| DDD intercept ( $\delta^D$ )                                     | −1.30<br>(2.00)                 | −3.05<br>(2.01)                    |
| Change in DDD per SD of baseline novelty ( $\theta^D$ )          | <b>2.94**</b><br>(1.47)         | <b>2.29*</b><br>(1.43)             |
| One-sided $p$ -value for $\theta^D > 0$                          | <b>.0312</b>                    | <b>.0636</b>                       |
| Joint pretrend $p$ -value for $\theta^D$                         | .5611                           | .4094                              |
| Observations                                                     | 79,578                          | 79,578                             |
| Closure-event clusters                                           | 18                              | 18                                 |
| $R^2$                                                            | .411                            | .410                               |
| Within $R^2$                                                     | .039                            | .038                               |
| Member-closure, relative-period and calendar-month fixed effects | Yes                             | Yes                                |
| Post-by-baseline-order interaction hierarchy                     | Yes                             | No                                 |

Notes: Estimates and standard errors are in percentage points. The dependent variable is novelty-seeking conditional on purchase. Baseline novelty-seeking is standardized over the 18,525 eligible customers described in Figure 4. Baseline orders are distinct transactions during relative periods  $-8$  through  $-1$ ; column (1) uses standardized  $\log(1 + \text{baseline orders})$  and the complete Post-by-baseline-order interaction hierarchy in equation (10), while column (2) deletes that hierarchy. In column (1),  $\delta^D$  is evaluated at mean baseline novelty and mean baseline order history; in column (2), it is evaluated at mean baseline novelty. Boldface marks the focal heterogeneity coefficient,  $\theta^D$ . The directional  $p$ -value applies only to  $\theta^D$  and tests the model-predicted alternative  $\theta^D > 0$ . Standard errors use CRV1 clustering by the 18 closure events. The joint pretrend row reports the  $p$ -value for the null that the event-study interaction coefficients in relative periods  $-4$ ,  $-3$  and  $-2$  are all zero.

planation as an important account of the novelty result. The notification contrast has the opposite sign from the one required. The assortment-timing contrast has the predicted sign, but its magnitude is small relative to the novelty DDD.

### 5.2.1 New-Product Notifications

Consumers receive push notifications from the company after they enter the “dormant” stage, which means they have no purchase in the past 30 days. The push notification is sent based on a algorithm known to the company but unknown to the researchers. However, we know that the algorithm is purely based on consumers’ history rather than personal information. One specific type of notification is a message that reminds consumers of the newly introduced products, which may trigger consumers’ novelty-seeking behavior. Since the notification is history-based, it’s possible that low- and high-intention consumers receive new product notifications with a different intensity, because low- and high-intention consumers are classified by our prediction algorithm in section 3.1 based on consumers’ history. If high-intention consumers receive an lower exposure to new product notification during the closure or after reopening, they will exhibit a lower

novelty-seeking level, consistent with our estimates in the main results. To generate the negative novelty DDD, the closure must differentially increase new-product exposure among treated low-intention customers relative to treated high-intention customers and to the corresponding control-group difference.

However, the notification records don't support the required evidence. We count unique new-product campaigns per customer-day and estimate the same treated-control, high-low contrast used for the main outcome. During the closure, the relative high-minus-low exposure change is +0.0041 campaigns per day (95% CI: 0.0019 to 0.0064; restricted wild-cluster  $p = 0.003$ ). After reopening, it remains positive at +0.0017 per day (95% CI: 0.0001 to 0.0032; restricted wild-cluster  $p = 0.047$ ), and the event-study leads do not reject a common pretrend ( $p = 0.500$ ). Extensive-margin measures yield the same pattern. Appendix C.1 reports the construction and full estimates.

The observed campaign pattern would, if anything, work against the negative novelty DDD. Under the natural assumption that new-product notifications weakly increase awareness or experimentation, relatively greater exposure among treated high-intention customers cannot explain their lower relative exploration.

### 5.2.2 Reopening Assortments

A supply-side explanation is that high- and low-intention consumers face different product assortments when they return after reopening. In particular, the two groups may return at different times, so differences in novelty-seeking could arise if the assortment changes systematically over the post-reopening period. This explanation therefore requires meaningful variation in product assortment with time since reopening. We test this implication by examining whether realized assortment at the treated stores changes over the first four weeks after reopening.

We construct three measures of realized assortment from product-level transactions. Core-product coverage measures the share of a store's regularly sold pre-closure products that continue to be sold in a given week. Products per 50 baskets measures the number of distinct products sold in standardized samples of 50 customer-day baskets, which prevents differences in transaction volume from mechanically generating differences in measured assortment. Pre-menu overlap measures the similarity between products sold in a given week and the store's pre-closure product set. We compare each measure across the four fixed seven-day periods following reopening.

Across the four post-reopening weeks, core-product coverage is statistically stable: relative to week 1, it changes by +0.59, -6.48, and -8.40 percentage points in weeks 2-4, and a joint restricted wild-cluster



test yields  $p = 0.382$ . The other two measures do evolve. Standardized product breadth changes by  $+0.30$ ,  $-0.70$ , and  $-1.66$  products per 50 baskets ( $p = 0.011$  jointly), while pre-menu overlap changes by  $-1.00$ ,  $-7.63$ , and  $-11.11$  percentage points ( $p = 0.018$  jointly). Thus, a test based only on assortment stability would not rule out the timing explanation.

We therefore combine the assortment paths with consumers' actual return timing. Among treated customers who return to the focal store within 28 days, high-intention customers return 2.87 days earlier than low-intention customers ( $SE = 0.29$ ); 53.6% of high-intention returners, compared with 36.9% of low-intention returners, return in week 1. For each returner and each of the four possible return weeks, we then construct a leave-one-customer-out realized product set and measure the share and count of products that the customer had never purchased by reopening. To isolate assortment timing, we subtract each customer's own four-week mean opportunity from the opportunity in the actual return week. The high-minus-low difference in this timing-induced deviation is only  $-0.10$  percentage points for the novel-product share ( $SE = 0.06$  percentage points; 95% CI:  $-0.22$  to  $+0.02$ ;  $p = 0.093$ ) and  $-0.28$  products for the novel-product count ( $SE = 0.38$ ; 95% CI:  $-1.09$  to  $+0.54$ ;  $p = 0.484$ ). The share estimate is about one-fortieth of the  $-4.15$ -percentage-point novelty DDD. Thus, although realized assortment changes after reopening and the two groups return at different times, the observed timing-induced opportunity difference weighs against this channel as a quantitatively important explanation for the main result. This is not a categorical rejection: recorded sales proxy product availability, and the exposure-share estimate and purchase-novelty DDD are not linked by a structural model.

The transaction data measure products actually sold rather than the exact assortment displayed or available to each individual consumer. Subject to this limitation, the customer-specific exposure calculation directly evaluates the observed assortment-timing channel rather than inferring it from store-level stability alone. Appendix C.2 reports the construction and additional analyses.

## 6 Robustness Checks

The robustness analysis addresses four distinct concerns about our main result. First, in our main specification (9), we adopt a binary classification of the purchase intention as high or low based on the predicted intention and a threshold of 0.5. We thus examine the robustness of replacing the binary prediction with the continuous prediction in the econometric specification. Second, our definition of novelty-seeking focus on

consumers' preference towards products that they never tried before. Another similar definition could focus on consumers' preference towards products that are newly introduced by the company. Third, our key finding and proposed mechanism could be related to the addictive nature of the coffee category. To generalize our findings, we restrict our attention to the non-coffee products. Last, we control for the impact of competition on our findings.

## 6.1 Continuous Predicted Purchase Probability

After replacing the binary prediction with the continuous prediction in (9), we find the sign of the main coefficient of interest does not depend on the 0.5 classification threshold. Replacing the binary indicator with the centered predicted-intention score yields a treatment-by-score coefficient of  $-0.0452$  for novelty-seeking ( $SE = 0.0276$ ;  $p = 0.120$ ). In economic terms, a 10-percentage-point increase in predicted intention therefore lowers the fitted treated-control post contrast by approximately 0.45 percentage points. The purchase-frequency coefficient is  $+0.0076$  ( $SE = 0.0051$ ;  $p = 0.152$ ), corresponding to a 0.08-point increase per day for a 10-point score change. The estimates match the binary results in sign but are imprecise under closure-clustered inference. Appendix Table W14 reports the complete specifications.

## 6.2 Alternative Outcome Definition: New-Product-Seeking

Our main definition of novelty-seeking focuses on consumers' preferences for products that they have not purchased before. An alternative definition focuses instead on consumers' preferences for products that are newly introduced. We therefore define new-product-seeking as the share of distinct products purchased within a purchasing window that are newly appear in the menu in the current or immediately preceding window. We use the same purchasing windows, treatment assignments, fixed effects, and closure-level clustering as in the main specification.

Using this alternative outcome, we find a larger and more significant reduction in new-product-seeking among consumers with high predicted purchase intention. The binary-intention DDD estimate is  $-4.17$  percentage points ( $SE = 0.89$ ;  $p < 0.001$ ), corresponding to 28.6% of the pre-period mean. The treated-control effect for high-intention consumers is  $-3.37$  percentage points ( $SE = 1.45$ ;  $p = 0.033$ ), compared with  $+0.80$  percentage points for low-intention consumers ( $SE = 1.13$ ;  $p = 0.487$ ). The corresponding high-minus-low pretrend test does not reject the null of parallel pretrends ( $p = 0.114$ ).

The result is also robust to using the continuous predicted purchase intention. A one-unit increase in the predicted-intention score reduces the treated-control post-period difference in new-product-seeking by 4.86 percentage points ( $SE = 1.56$ ;  $p = 0.006$ ). Equivalently, a 10-percentage-point increase in predicted intention is associated with a 0.49-percentage-point larger reduction. The corresponding pretrend test does not reject ( $p = 0.377$ ). Appendix D.2 reports the complete estimates. Overall, these results show that our main finding is robust to defining novelty based on whether a product is newly introduced rather than solely on whether the consumer has purchased it before.

### 6.3 Effect on Non-Coffee Products

Our main analysis includes all products sold, where nearly a half is coffee products, for which purchasing behavior may be shaped by habit or the addictive nature of caffeine. One may question that our main finding is based on the specific category of coffee products. To examine whether our findings generalize beyond coffee, we re-estimate the novelty-seeking specification using only non-coffee products. Based on the commodity-description fields, we consider three increasingly restrictive product groups: all non-coffee products, non-coffee consumables, and non-coffee beverages. Within each group, novelty-seeking is defined as the share of distinct products purchased in a purchasing window that the consumer had not purchased previously. The outcome is therefore observed only for windows in which the consumer purchases at least one product from the corresponding group.

Across all three product groups, the estimated effect remains negative. For all non-coffee products, the DDD estimate is  $-4.13$  percentage points ( $SE = 3.40$ ; restricted wild-cluster  $p = 0.225$ ), close to the  $-4.15$ -percentage-point estimate for our main novelty-seeking outcome. Restricting the sample to non-coffee consumables yields an estimate of  $-4.11$  percentage points, while restricting it further to non-coffee beverages yields an estimate of  $-6.26$  percentage points. Although all three estimates are similar in direction to the main result, they are imprecisely estimated, because of a smaller number of observations.

The diagnostic tests, however, suggest caution in interpreting these estimates as additional causal evidence. The all-non-coffee and non-coffee-consumables specifications reject the parallel triple-trends assumption ( $p = 0.006$  in both cases). For non-coffee beverages, the novelty-seeking pretrend test does not reject ( $p = 0.226$ ), but high-predicted-intention treated consumers become less likely to make a qualifying purchase after treatment ( $DDD = -2.24$  percentage points;  $SE = 0.90$ ;  $p = 0.023$ ). Because novelty-seeking can be measured only when such a purchase occurs, this differential change may affect the composition of

consumers for whom the outcome is observed. Appendix D.3 reports the complete estimates and diagnostic tests. We therefore view this exercise as evidence that the negative pattern is not unique to coffee products, while the diagnostic results prevent us from assigning these estimates the same causal interpretation as our main findings.

## 6.4 Accounting for Local Competition

Local competition may affect consumers' product choices after reopening and could therefore contribute to our main findings. To account for this possibility, we supplement the transaction data with store-address data that report the number of potentially competing stores located within 500 and 1,500 meters of each store. We use the 500-meter count as our primary measure of local competition and assign it to each consumer through the consumer's preferred store in the pre-closure period. Because these counts are time invariant, we allow their effect to vary after reopening by interacting local competition with the post-period indicator, treatment status, and predicted purchase intention, while retaining the corresponding lower-order terms. Appendix D.4 reports the data construction, full specification, alternative measures, and diagnostic tests. This specification therefore allows the post-reopening treated-control difference between high- and low-intention consumers to vary systematically with the level of competition surrounding their preferred store. The reported DDD at the mean competitor count can consequently be interpreted as the treatment effect evaluated at the average observed level of local competition.

The negative novelty-seeking effect remains after accounting for local competition in this way. In the primary specification using the number of nearby establishments within 500 meters, the fitted DDD at the mean level of competition is  $-3.33$  percentage points (95% CI:  $-6.35$  to  $-0.31$ ), compared with  $-4.15$  percentage points in the main specification. The fitted DDD also remains negative at competitor counts of 2, 5, and 13, with estimates of  $-5.03$ ,  $-4.39$ , and  $-2.70$  percentage points, respectively. A one-standard-deviation increase in local competition changes the DDD by an estimated  $+3.00$  percentage points, although this relationship is imprecisely estimated (95% CI:  $-5.15$  to  $11.14$ ). Using the logarithm of the competitor count or the alternative 1,500-meter measure produces similarly negative fitted DDD estimates at their respective means. As an additional check, we include preferred-store-by-event-by-post fixed effects to absorb post-reopening shocks shared by consumers associated with the same preferred store. The resulting DDD remains negative at  $-4.45$  percentage points (95% CI:  $-8.07$  to  $-0.84$ ).

These results indicate that our main finding is not explained by observed differences in the local com-

petitive environment. However, the auxiliary data measure only the number of nearby establishments: they do not identify their brands, operating status, or whether they temporarily closed during the study period. In addition, the competition-interacted event-study leads reject equality to zero ( $p = 0.002$ ), which limits a causal interpretation of how competition moderates the treatment effect. We therefore interpret this exercise as a robustness check showing that the negative novelty-seeking effect persists after allowing post-reopening behavior to vary with observed local competition, rather than as evidence identifying a separate competition mechanism.

## 7 Conclusion

This paper studies how consumers respond when a temporary store closure is likely to interrupt a purchase. Using panel data from a large digital coffee chain, we estimate a triple-differences design that compares high- and low-predicted-intention post-reopening responses. Purchase frequency is 0.58 percentage points per day higher for high- than for low-predicted-intention customers after reopening. Because the corresponding pretrend test rejects equality, this estimate is descriptive evidence of recovery rather than a clean causal rebound.

The product-choice result is more credible under the design’s pretrend diagnostics. Novelty-seeking responds 4.15 percentage points less for high- than for low-predicted-intention customers. The direct high-group effect is a statistically insignificant 1.02-point decline. Under parallel triple trends and comparable general closure effects, this DDD identifies the incremental response associated with interrupting a likely purchase. An appendix robustness check based on new-product-seeking yields the same qualitative conclusion when exploration is defined relative to products that have appeared recently in the transaction data. Consistent with familiar-choice salience, the interruption-related narrowing of product exploration is stronger among consumers with lower baseline novelty-seeking.

A goal-persistence mechanism is one interpretation: an interrupted likely purchase may make familiar products more salient upon return. The main DDD does not isolate this psychological channel from other reasons that an interruption may reduce exploration.

Our findings contribute to the marketing literature in several ways. First, we show how a temporary constraint interacts with ex-ante purchase intention to shape post-reopening behavior; persistent purchase goals are one possible account of the product-choice response rather than a separately identified mechanism.

Second, we show that supply-side shocks can have distinct effects on the level and composition of demand, and that these effects depend critically on whether the constraint is likely to bind. By focusing on consumers for whom the closure likely interrupts a purchase, we uncover behavioral responses that can be masked in aggregate analyses. Furthermore, our study contributes to a broader understanding of product discovery by showing how temporary interruptions can change consumer choice even when demand levels recover quickly.

From a managerial perspective, purchase-day frequency can recover while product discovery weakens. The novelty-seeking estimate is a relative effect, and the alternative new-product-seeking definition strengthens the evidence that exploration narrows after a likely purchase is interrupted. Firms should therefore monitor product discovery alongside post-disruption sales recovery. More generally, demand recovery is incomplete when customers return but narrow the set of products they consider.

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## Web Appendix

### A Data Construction and Measurement

This appendix records construction details that are needed to interpret the empirical design but are too mechanical for the main text. The unit of assignment is a customer-closure consumer. A customer enters through a closure event for which her pre-closure purchase history identifies a meaningful focal store; the outcomes then aggregate her purchases across the observed chain.

#### A.1 Closure Registry and Customer-consumer Sample

The raw data contain 7,312,498 orders, represented by 10,631,250 product-line records, from 779,744 customers at 260 stores between June 1, 2020 and December 31, 2021. We aggregate these orders to a store-day panel and define a candidate closure as a consecutive spell in which a store records no transactions. This procedure identifies 101 zero-transaction spells.

We first remove 60 spells at stores located in universities or colleges. These spells occur during summer periods in which COVID-19 policies may have kept students away from campus. Because the data cover purchases made in this city, we do not observe whether students continued to buy from the chain elsewhere. A campus store with no local transactions may therefore reflect the temporary absence of its customers rather than a supply disruption. Removing these spells avoids treating a seasonal loss of the local customer population as a store closure and leaves 41 non-campus candidates.

We next exclude seven spells lasting 30 days or longer. The analysis is designed around short disruptions followed by a clearly observed reopening; longer spells are more likely to reflect seasonal shutdowns or persistent changes in store operations. They also produce much longer pre- and post-event windows, which are difficult to compare within the 19-month data period. Requiring closures to last strictly fewer than 30 days leaves 34 candidates.

The remaining events must provide enough support for the treated-control comparison. We require at least one treated and one control customer and at least 50 customers in each group. These restrictions remove events for which the DDD is undefined or rests on a very small group. We also require the control purchase rate during the closure to be at least twice the treated rate. This screen is intended to distinguish a localized access disruption from a broader contraction in demand that affects treated and control customers alike.

Together, the support and purchase-rate screens remove 12 events—eight for a low control purchase rate, two for insufficient group size and two for having no control customers—leaving 22 closures. Because the purchase-rate screen uses behavior realized during the closure, the full-registry sensitivity analysis remains important.

The maintained headline sample excludes four additional events around the Lunar New Year period. Travel, holiday schedules and COVID-19-related mobility changes during this period can shift customer traffic independently of a store closure, and these events have weaker treatment-control fit. This restriction leaves 18 closures lasting 10 to 29 days. It is not generated by a coded ex-ante rule, so we retain the 22-event full-registry analysis as a sensitivity check. The novelty DDD keeps its sign in that broader registry, while the purchase-frequency estimate is less stable across broader sample definitions; the archived sensitivity analysis records those comparisons without changing the 18-event headline registry.

For each retained closure, treated customers are those whose preferred pre-closure store is the store that closes. Control customers are those whose preferred store is one of the matched stores that remains open. We define the preferred store from pre-closure purchases and require at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. These restrictions ensure that the closing store was an economically meaningful source of access for treated customers. The final sample contains 40,148 customer-closure customers: 7,390 treated and 32,758 controls. Of the treated customers, 2,041, or 27.6%, have high predicted purchase intention during the closure window.

## A.2 Outcome Timing and Variable Definitions

Each closure event defines relative periods  $t = -4, \dots, 4$ . Every relative period has the same length as the closure, denoted  $L_e$ , and period 0 is the closure window itself. The main post-reopening regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This convention produces 321,184 customer-consumer-period observations for purchase frequency. Customers make at least one purchase in 105,854 of these rows, so novelty-seeking is defined in those rows. The consumer fixed effects then remove 6,210 rows belonging to customers with only one product-choice period, leaving 99,644 observations in each main product-choice regression. Table 2 reports pre-period purchase probabilities across the four DDD cells.

Table W1: Key variables in the empirical design

| Variable                          | Construction                                                                                                                                                                                 | Interpretation                                                                                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Treated                           | Preferred pre-closure store is the closed store.                                                                                                                                             | Exposure to the local store closure. This is not the same as having a purchase blocked.                                               |
| Control                           | Preferred pre-closure store is a matched store that remains open.                                                                                                                            | Counterfactual local-store path for treated customers.                                                                                |
| High predicted purchase intention | Ex-ante predicted probability of at least one purchase during the closure-length window is at least .5.                                                                                      | Predicted closure-window purchase intention. It is defined before observing post-reopening outcomes.                                  |
| Interrupted purchase              | Treated $\times$ High predicted purchase intention.                                                                                                                                          | The closure is predicted to bind: the customer is exposed to the closure and likely would have purchased during the closure window.   |
| Post                              | Relative periods 1 through 4.                                                                                                                                                                | Post-reopening windows. Period 0 is the closure window and is omitted from the main DDD regressions.                                  |
| Purchase frequency                | Days with at least one purchase at any chain store in the relative-period window, divided by $L_e$ .                                                                                         | Chain-wide extensive-margin purchase intensity, normalized for closure length; multiple orders on one day count once.                 |
| Novelty-seeking                   | Share of distinct products purchased anywhere in the chain in the window that the customer had not previously purchased anywhere in the chain.                                               | Chain-wide product exploration relative to the customer's own observed history, conditional on purchase.                              |
| Baseline novelty-seeking          | consumer-level mean novelty-seeking over relative periods $-8$ through $-1$ , restricted to customers with at least five orders over those periods and standardized over eligible customers. | Revealed baseline orientation toward familiar versus novel products. Relative periods are closure-length windows, not calendar weeks. |
| New-product-seeking               | Share of distinct products purchased anywhere in the chain whose first observed chain sale occurs in the current or immediately previous event window.                                       | Adoption of recently appearing products, conditional on purchase. First observed sale need not equal the formal catalog launch date.  |

Notes: The main text uses “high predicted purchase intention” for customers with a score above the threshold. A likely interrupted purchase occurs only for treated high-predicted-intention customers, because controls retain access to their focal store.

### A.3 Purchase Probability and the Novelty-Seeking Sample

Novelty-seeking is defined only in customer-consumer windows containing a purchase. Figure 3 in the main text plots the realized probability of at least one purchase in each DDD cell and relative period. This is also the probability that a window enters the novelty-seeking sample. Purchase probability is much higher among high-predicted-intention customers, as expected from a classifier designed to predict purchasing. The informative comparison is treated versus control within each intention stratum, where the paths are close. High-intention purchase probability falls from 0.66–0.85 across the pre-periods to roughly 0.56–0.61 after reopening, whereas low-intention purchase probability remains near 0.17–0.27. The conditional novelty

regressions thus compare purchasing windows drawn from a population whose composition changes over time. Similar treated and control purchase-probability paths reduce concern about differential sample entry, but they do not establish that the observed purchasers are compositionally identical. The main estimates therefore retain a conditional-on-purchase interpretation.

#### A.4 Predicted Purchase-intention Classifier

The classifier separates closure exposure from likely purchase interruption. It predicts whether a customer would purchase at least once anywhere in the chain in a window with the same length as the closure. This target depends mechanically on window length, so the implementation trains one XGBoost model for each observed closure duration. Training uses pre-closure relative periods  $-4$  through  $-2$ , validation uses the holdout pre-period  $-1$ , and an additional audit uses control customers in period 0 because their matched focal stores remain open. The period-0 audit is a validation exercise; the DDD uses the ex-ante predicted score.

Table W2: Classifier implementation details

| Component           | Implementation                                                                                                                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prediction target   | Indicator for at least one purchase anywhere in the chain in a closure-length window.                                                                                                                          |
| Model family        | XGBoost binary classifier, estimated separately by closure duration.                                                                                                                                           |
| Training windows    | Relative periods $-4$ , $-3$ and $-2$ .                                                                                                                                                                        |
| Holdout validation  | Relative period $-1$ , separately audited for treated and control customers.                                                                                                                                   |
| Additional audit    | Control customers in period 0, where the matched control store remains open and actual purchase behavior is observed.                                                                                          |
| Scored sample       | Treated and control customer-closure customers in the ex-ante period-0 panel.                                                                                                                                  |
| Classification rule | High predicted purchase intention equals one if the predicted period-0 purchase probability is at least .5.                                                                                                    |
| Feature families    | Recent purchase frequency, recency, total purchase history, preferred-store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history. |

Notes: The performance table in the main text and the diagnostics below use the exact final 18-event cohort. The implementation table emphasizes timing and information sets, because these are the key design concerns for interpreting the classifier as ex ante. The .5 cutoff is fixed in the classifier configuration and is not selected using the post-reopening outcomes.

The classifier discriminates reasonably well in every production duration: period-0 control ROC-AUC ranges from 0.739 to 0.794. Calibration is consistently weaker, with duration-specific slopes from 0.331 to 0.592. Changing the cutoff produces the expected sensitivity-specificity tradeoff rather than revealing a uniquely preferred threshold. These diagnostics motivate cautious language for the binary group and the continuous-score robustness specification; they do not turn predicted intention into a direct measure of

Table W3: Purchase-intention classification at the .5 cutoff

| Evaluation sample | TN     | FP    | FN    | TP    | Accuracy | Benchmark | Sensitivity | Specificity | PPV  | NPV  | F1   |
|-------------------|--------|-------|-------|-------|----------|-----------|-------------|-------------|------|------|------|
| Control, $t = 0$  | 20,045 | 4,111 | 3,889 | 4,713 | .756     | .737      | .548        | .830        | .534 | .838 | .541 |
| Control, $t = -1$ | 19,456 | 3,214 | 4,321 | 5,767 | .770     | .692      | .572        | .858        | .642 | .818 | .605 |
| Treated, $t = -1$ | 4,368  | 698   | 971   | 1,353 | .774     | .686      | .582        | .862        | .660 | .818 | .619 |

Notes:  $t = 0$  is the closure window and  $t = -1$  is the holdout pre-period. TN, FP, FN and TP denote true negatives, false positives, false negatives and true positives. Benchmark is the accuracy of predicting no purchase for every consumer. Sensitivity is  $TP/(TP+FN)$ , specificity is  $TN/(TN+FP)$ , PPV is  $TP/(TP+FP)$ , and NPV is  $TN/(TN+FN)$ . All rows use the exact final cohort.

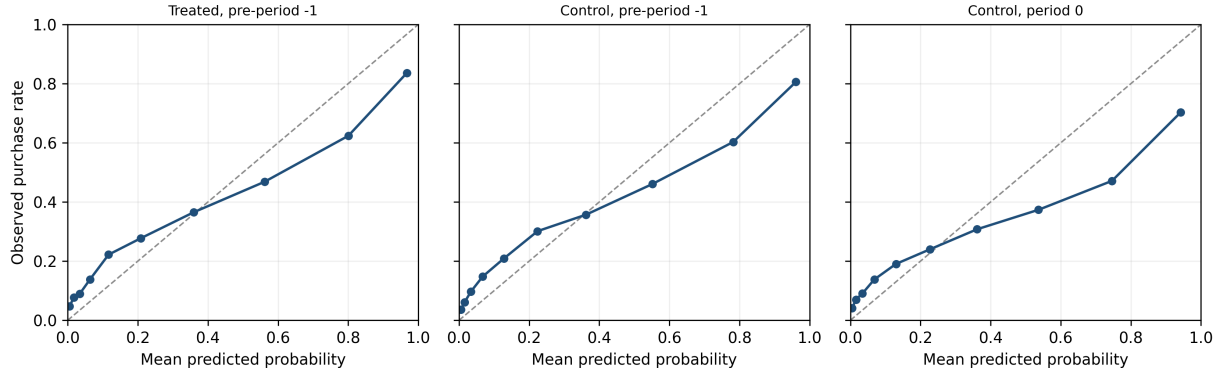


Figure W1: Calibration of predicted purchase intention

Notes: Each point is one decile of predicted purchase probability in the exact final cohort. The horizontal coordinate is the decile's mean prediction and the vertical coordinate is its observed purchase rate. The dashed 45-degree line denotes perfect calibration. The flatter empirical paths show that the raw probabilities are too dispersed, although they continue to rank purchase intention.

Table W4: Period-0 control performance by closure duration

| Duration | $N$   | Prevalence | ROC-AUC | PR-AUC | Brier | Calibration slope |
|----------|-------|------------|---------|--------|-------|-------------------|
| 10 days  | 1,244 | .210       | .774    | .501   | .157  | .461              |
| 11 days  | 5,020 | .246       | .783    | .585   | .152  | .548              |
| 12 days  | 3,164 | .196       | .786    | .508   | .154  | .432              |
| 13 days  | 1,662 | .238       | .780    | .525   | .179  | .388              |
| 14 days  | 5,232 | .230       | .783    | .555   | .148  | .592              |
| 15 days  | 551   | .376       | .754    | .686   | .210  | .374              |
| 16 days  | 2,416 | .300       | .794    | .646   | .176  | .448              |
| 17 days  | 2,783 | .304       | .789    | .627   | .183  | .478              |
| 18 days  | 3,176 | .224       | .786    | .569   | .155  | .537              |
| 26 days  | 4,645 | .299       | .761    | .592   | .208  | .494              |
| 28 days  | 1,891 | .347       | .791    | .689   | .199  | .489              |
| 29 days  | 974   | .359       | .739    | .611   | .232  | .331              |

Notes: This table evaluates control customers in period 0 separately for each duration-specific classifier. Duration 25 does not occur in the final 18-event registry and is excluded even though a sensitivity-run artifact exists in the classifier output directory.

Definitions follow Table 1.

psychological intention.

Table W5: Predictive tradeoffs across classification cutoffs

| Cutoff | Predicted high | Accuracy | Sensitivity | Specificity | PPV  |
|--------|----------------|----------|-------------|-------------|------|
| .30    | .391           | .712     | .695        | .718        | .468 |
| .40    | .325           | .738     | .620        | .780        | .502 |
| .50    | .269           | .756     | .548        | .830        | .534 |
| .60    | .216           | .770     | .474        | .875        | .575 |
| .70    | .170           | .777     | .398        | .911        | .615 |

Notes: The evaluation sample is the exact-cohort period-0 controls. Predicted high is the share classified above the stated probability cutoff. These are predictive tradeoffs, not DDD estimates. The main .5 rule is configured before estimation; the continuous-score DDD avoids discretizing the score.

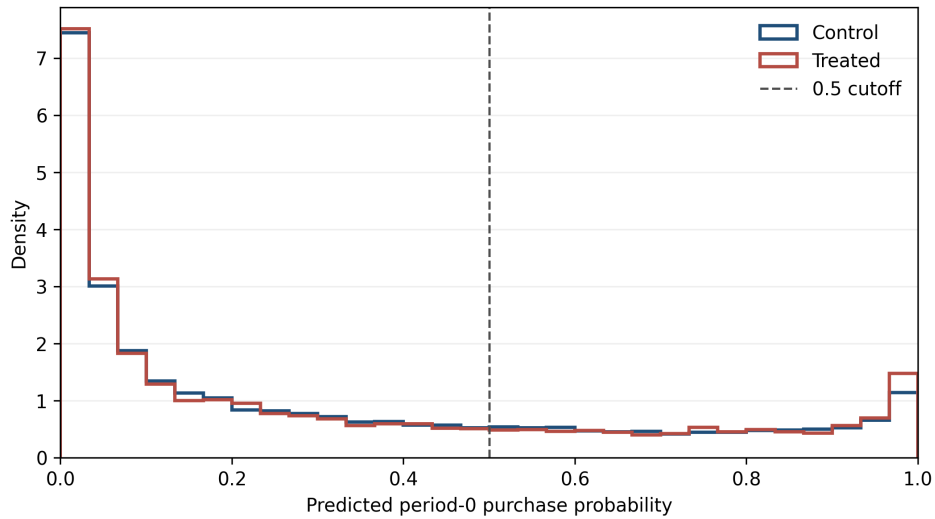


Figure W2: Overlap in predicted period-0 purchase intention

Notes: The figure shows density histograms of the ex-ante score in the exact final cohort. Mean predicted intention is 0.307 for controls and 0.313 for treated customers; the median is 0.172 and 0.174, respectively. The shares at or above 0.5 are 26.9% and 27.6%. The similar distributions indicate strong score overlap across treatment arms, although overlap does not validate calibration for the unobserved treated period-0 counterfactual.

## B Identification Diagnostics

This appendix reports the formal diagnostics behind the identification discussion in Section 4.2.1. The main design has three dimensions: treatment exposure, post-reopening timing and predicted purchase intention. The key identifying restriction is therefore a triple-difference restriction: absent the closure, the treated-control gap for high-predicted-intention customers would have changed like the corresponding gap for low-predicted-intention customers.

### B.1 Pretrend Tests

For the main novelty-seeking result, we estimate the event-study analog of equation (9):

$$Y_{iet}^N = \alpha_{ie} + \lambda_t + \gamma_m + \sum_{\ell \in \{-4, -3, -2, 1, 2, 3, 4\}} \left[ \delta_\ell^B \text{Treated}_{ie} + \beta_\ell \text{High}_{ie} + \delta_\ell^D (\text{Treated}_{ie} \times \text{High}_{ie}) \right] \mathbb{I}\{t = \ell\} + \varepsilon_{iet}. \quad (11)$$

Period  $-1$  is the omitted reference period, and period  $0$  is the closure window and is excluded. The coefficient  $\delta_\ell^D$  is the high-minus-low DDD at relative period  $\ell$ . The pretrend test is the joint null  $H_0 : \delta_{-4}^D = \delta_{-3}^D = \delta_{-2}^D = 0$ .

Table W6: Novelty-seeking DDD pretrend coefficients and joint test

| <i>Panel A: Pre-period high-minus-low DDD coefficients</i> |                |              |               |
|------------------------------------------------------------|----------------|--------------|---------------|
| Relative period                                            | DDD estimate   | SE           | 95% CI        |
| −4                                                         | 2.23           | 2.73         | [−3.52, 7.98] |
| −3                                                         | 2.64           | 3.44         | [−4.61, 9.90] |
| −2                                                         | 1.60           | 1.98         | [−2.57, 5.77] |
| <i>Panel B: Joint test</i>                                 |                |              |               |
| Null hypothesis                                            | Wald statistic | Restrictions | p-value       |
| $\delta_{-4}^D = \delta_{-3}^D = \delta_{-2}^D = 0$        | .837           | 3            | .8407         |

Notes: Coefficients and standard errors are in percentage points. The dependent variable is novelty-seeking, defined only in windows with a purchase. The regression uses 99,644 observations and includes customer-closure, relative-period and calendar-month fixed effects. Period  $-1$  is the omitted reference, and period  $0$  is excluded. Standard errors are clustered by closure event (18 clusters). The joint Wald test uses the closure-clustered covariance matrix.

The three novelty-seeking lead estimates are positive, between 1.60 and 2.64 percentage points, and each confidence interval includes zero. The joint test does not reject that all three are zero. This is evidence against a pronounced differential pretrend in the observed pre-closure windows, but it does not prove the identifying

assumption. Replacing the binary high-intention indicator with the continuous predicted-intention score likewise does not reject a differential pretrend (Wald statistic = 1.781,  $p = 0.619$ ). For transparency about the other outcomes, the purchase-frequency high-minus-low test rejects ( $p = 0.035$ ), which is why that result is treated as descriptive, while the new-product-seeking robustness outcome does not reject ( $p = 0.114$ ).

Figure W3 shows the novelty high-minus-low coefficient path. The omitted reference period is  $-1$ , and period  $0$  is the closure window and is excluded from estimation. Each displayed 95% confidence interval includes zero. The post-reopening point estimates are negative, but individually imprecise; the figure is therefore evidence about the timing and uncertainty of the response, not a claim that every post-period coefficient differs from zero.

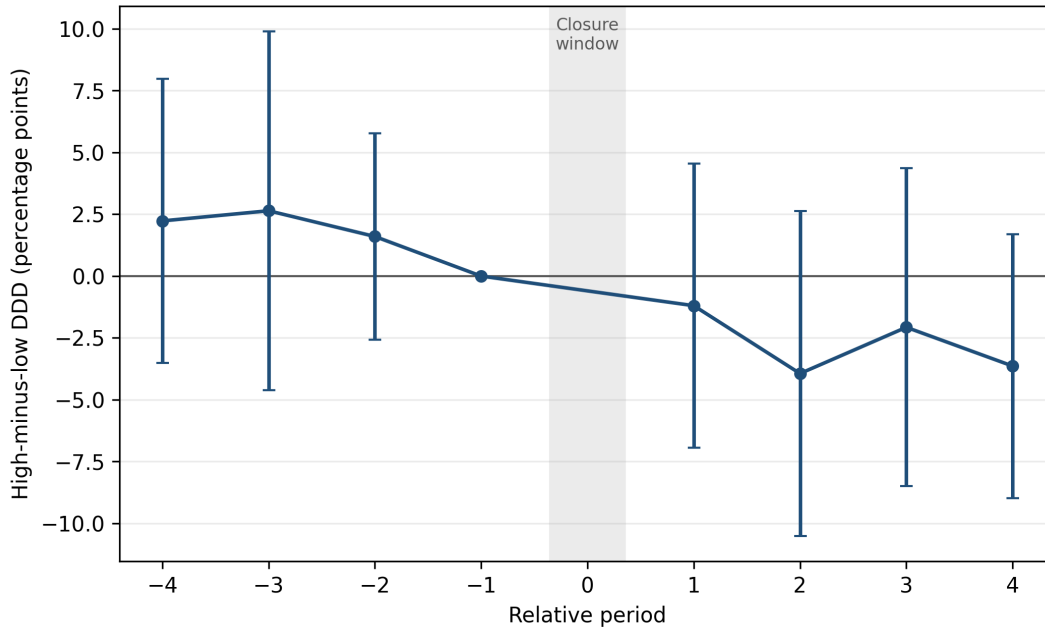


Figure W3: Event-study path of the novelty-seeking DDD

Notes: Points are the event-study coefficients on Treated  $\times$  High predicted intention relative to period  $-1$ . Vertical bars use the confidence limits reported by the closure-event-clustered estimator (18 clusters). Period  $0$  is the closure window and is omitted. Novelty-seeking is conditional on a purchase in the window.

## B.2 Common-Support Sensitivity

The identifying comparison can be strained when high- and low-predicted-intention customers have different pre-period demand. We therefore construct a deterministic common-support sample separately within treated and control customers. Coarsened cells use mean and most recent pre-period purchase fre-



quency and purchase recency; the novelty match additionally uses pre-period purchase intention because novelty is defined only in purchasing windows. Within each cell, we retain equal numbers of high- and low-predicted-intention customers.

Table W7: Common-support sensitivity of the DDD estimates

| Outcome            | Full sample                         |              | Matched common support              |              |
|--------------------|-------------------------------------|--------------|-------------------------------------|--------------|
|                    | DDD                                 | Observations | DDD                                 | Observations |
| Purchase frequency | <b>.0058</b><br>(.0032)<br>[.0860]  | 321,184      | <b>-.0041</b><br>(.0027)<br>[.1561] | 101,072      |
| Novelty-seeking    | <b>-.0415</b><br>(.0184)<br>[.0380] | 99,644       | <b>-.0211</b><br>(.0195)<br>[.2955] | 43,655       |

Notes: The table reports the high-minus-low DDD, with all DDD coefficients shown in boldface. The matched sample imposes common support between high- and low-predicted-intention customers separately within treated and control groups using the coarsened cells described in the text. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. The matched estimates are sensitivity diagnostics rather than preferred estimates.

The support restriction retains 12,634 of 40,148 customers for purchase frequency and 10,956 customers for novelty-seeking before the product-choice singleton restriction. The purchase-frequency DDD changes sign and is imprecise ( $p = 0.156$ ); its matched-sample DDD pretrend also does not reject at 5% but is borderline ( $p = 0.069$ ). The novelty DDD is about half the full-sample magnitude and is imprecise ( $p = 0.295$ ), while its matched pretrend does not reject ( $p = 0.871$ ). The common-support exercise therefore does not provide affirmative robustness: it shows that the headline estimates, especially purchase frequency, are sensitive to restricting the comparison to closely aligned pre-period demand cells.

### B.3 Dynamics of Baseline-Novelty Heterogeneity

The collapsed heterogeneity estimate in Table 4 compares all four post-reopening periods with all four pre-closure periods. Figure W4 replaces the collapsed post indicator with relative-period interactions, retaining the same fixed effects, baseline-order adjustment and closure-event clustering. The omitted reference period is  $-1$ .

The three lead coefficients are small relative to their confidence intervals, and their joint test does not reject equality to zero ( $p = 0.561$ ). The post-period gradients are positive in periods 1, 3 and 4 and near zero in period 2. The largest estimate occurs immediately after reopening, but every period-specific 95% confidence interval includes zero. The figure therefore supports the absence of a pronounced differential

pretrend while also showing that the timing of the mechanism result is imprecisely estimated.

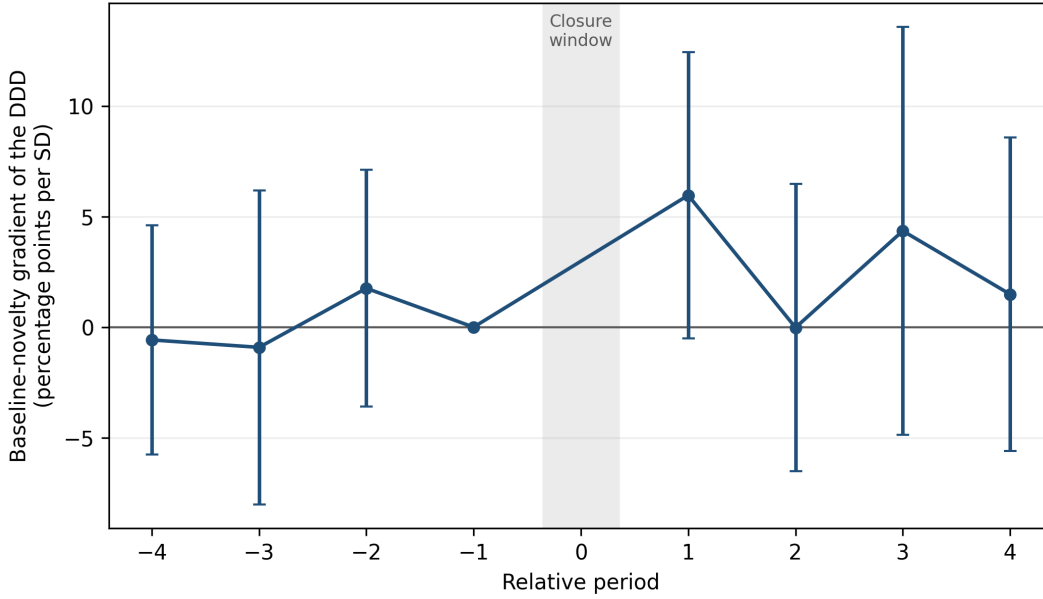


Figure W4: Event-study path of the baseline-novelty gradient

Notes: Points are coefficients on Relative period  $\times$  Treated  $\times$  High predicted intention  $\times$  standardized baseline novelty-seeking. Vertical bars are 95% confidence intervals from CRV1 standard errors clustered by closure event (18 clusters). Relative period  $-1$  is the omitted reference period; period 0 is the closure window and is excluded. The specification includes the full lower-order interaction hierarchy for baseline novelty and the parallel hierarchy for standardized log baseline orders, as in equation (10).

## C Alternative Explanations

### C.1 New-Product Notification Exposure

This section reports the construction and estimates behind Section 5.2.1. The raw files contain 56,724,521 push records. We retain the 40,148 consumers in the 18-closure analysis and map notifications into the four event-length periods before closure, the closure window and the four event-length periods after reopening. This produces a balanced panel of 361,332 member-event-period observations. Of the 912,911 push records mapped into these windows, 125,918 have `trigger_tag=3`, the platform classification for a new-product notification.

The primary measure counts unique member-policy-date notification campaigns and divides by the number of calendar days in the event window. All 125,918 new-product records are unique at this grain, so the campaign and raw-record measures coincide. Supporting outcomes indicate whether a member receives any new-product notification, count distinct notification days and measure the share of all campaigns clas-

sified as new-product notifications. The source files record targeting entries; they do not verify delivery, impression, opening or reading.

For exposure outcome  $N_{iet}$ , we estimate

$$N_{iet} = \beta_1(\text{After}_{et} \times \text{Treated}_i) + \beta_2(\text{After}_{et} \times \text{High}_i) + \beta_3(\text{After}_{et} \times \text{Treated}_i \times \text{High}_i) + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}, \quad (12)$$

where  $\text{After}_{et}$  denotes either the closure window or the post-reopening periods. The closure specification compares period 0 with all four pre-periods; a sensitivity check compares it only with period  $-1$ . The post specification omits period 0 and compares the four post-periods with the four pre-periods, as in the main outcome regression. Standard errors are clustered by closure event. Because there are 18 clusters, Table W8 also reports restricted wild-cluster  $p$ -values for  $\beta_3$ .

Table W8: Differential exposure to new-product notifications

|                                       | During versus all pre-periods<br>(1) | During versus period $-1$<br>(2) | Post versus pre-periods<br>(3) |
|---------------------------------------|--------------------------------------|----------------------------------|--------------------------------|
| New-product campaigns per day         | 0.0041<br>(0.0011)                   | 0.0043<br>(0.0013)               | 0.0017<br>(0.0007)             |
| Restricted wild-cluster $p$ -value    | [0.0032]                             | [0.0111]                         | [0.0469]                       |
| Any new-product notification          | 0.0422<br>(0.0100)                   | 0.0362<br>(0.0110)               | 0.0144<br>(0.0061)             |
| Restricted wild-cluster $p$ -value    | [0.0010]                             | [0.0057]                         | [0.0399]                       |
| New-product notification days per day | 0.0041<br>(0.0011)                   | 0.0043<br>(0.0013)               | 0.0017<br>(0.0007)             |
| Restricted wild-cluster $p$ -value    | [0.0031]                             | [0.0111]                         | [0.0471]                       |
| All campaigns per day                 | 0.0230<br>(0.0087)                   | 0.0143<br>(0.0078)               | 0.0113<br>(0.0058)             |
| Restricted wild-cluster $p$ -value    | [0.0823]                             | [0.1648]                         | [0.1104]                       |
| New-product share of campaigns        | 0.0020<br>(0.0095)                   | $-0.0014$<br>(0.0057)            | $-0.0005$<br>(0.0058)          |
| Restricted wild-cluster $p$ -value    | [0.8505]                             | [0.8080]                         | [0.9406]                       |
| Observations, primary outcome         | 200,740                              | 80,296                           | 321,184                        |
| Closure-event clusters                | 18                                   | 18                               | 18                             |

Notes: Entries are coefficients on  $\text{After} \times \text{Treated} \times \text{High}$  from equation (12). A negative coefficient is required for the proposed notification explanation. Standard errors clustered by closure event are in parentheses; restricted wild-cluster  $p$ -values are in brackets. All specifications include member-closure, relative-period and calendar-month fixed effects. The campaign-share regressions are conditional on receiving at least one campaign and use 107,606, 39,628 and 183,480 observations in columns (1)–(3), respectively.

Table W9 reports the dynamic specification, which replaces  $\text{After}$  with relative-period indicators and omits period  $-1$ . The three lead estimates are individually small, and their joint test does not reject equality to zero ( $p = 0.500$ ). The coefficient is largest during closure, remains positive in the first reopening period and then declines. This timing is consistent with high-intention treated consumers becoming newly eligible

for inactivity-triggered campaigns when a likely purchase is interrupted. It is inconsistent with the negative differential exposure needed to explain the novelty DDD.

Table W9: Event-study path of new-product notification exposure

| Relative period                | Coefficient | Standard error | <i>p</i> -value |
|--------------------------------|-------------|----------------|-----------------|
| −4                             | 0.0007      | (0.0010)       | [0.466]         |
| −3                             | −0.0002     | (0.0012)       | [0.878]         |
| −2                             | 0.0002      | (0.0009)       | [0.800]         |
| 0                              | 0.0043      | (0.0013)       | [0.004]         |
| 1                              | 0.0031      | (0.0013)       | [0.036]         |
| 2                              | 0.0015      | (0.0011)       | [0.185]         |
| 3                              | 0.0022      | (0.0011)       | [0.064]         |
| 4                              | 0.0003      | (0.0008)       | [0.712]         |
| Joint pretrend <i>p</i> -value |             | 0.500          |                 |

Notes: The outcome is unique new-product notification campaigns per consumer-day. Coefficients are relative-period  $\times$  Treated  $\times$  High interactions relative to period −1. Standard errors use CRV1 clustering by the 18 closure events. Period 0 is the closure window.

The inactivity pattern in the records also clarifies why absolute high-low levels are not informative. Among treated consumers, the median new-product notification occurs 121 days after the last purchase for low-intention customers before closure and 129 days during closure. The corresponding medians for high-intention customers are 56 and 40 days. Low-intention consumers therefore receive more notifications in levels because they are ordinarily more dormant. The DDD removes this persistent gap and shows that the relative exposure increase occurs among treated high-intention consumers instead.

## C.2 Reopening Assortment

This section reports the construction and estimates behind the assortment test in Section 5.2.2. The alternative explanation is that high- and low-intention consumers may return at different times after a treated store reopens and therefore encounter different product assortments. For differences in return timing to generate differences in product choice, however, the assortment available at the store must itself vary systematically with time since reopening. We therefore focus first on the evolution of realized assortment at the 18 treated stores during the first four weeks after reopening.

For each closure, we construct four fixed seven-day periods beginning immediately after reopening. Fixed weeks provide a common measure of time since reopening across events even though the preceding closures last between 10 and 29 days. We construct three measures from product-level transactions. *Core-*

*product coverage* defines a store’s core set as products sold in at least three of the four weeks before closure and measures the share of that set sold in each post-reopening week. *Products per 50 baskets* draws 50 customer-day baskets without replacement, counts the number of distinct products, and averages this count over 100 draws. This rarefaction holds observed basket volume fixed; store-weeks with fewer than 50 baskets are missing. *Pre-menu overlap* is the Jaccard similarity between products sold in a store-week and the union of products sold in that store’s four pre-closure weeks. These measures capture different dimensions of the realized product assortment while avoiding a mechanical interpretation of lower transaction volume as lower product availability.

**Assortment evolution after reopening.** For assortment outcome  $A_{et}$  at treated store associated with closure event  $e$  in post-reopening week  $t$ , we estimate

$$A_{et} = \alpha_e + \sum_{k=2}^4 \beta_k \mathbb{1}\{t = k\} + \varepsilon_{et}, \quad (13)$$

where week 1 is the omitted category and  $\alpha_e$  denotes closure-event fixed effects. The coefficients  $\beta_k$  therefore measure how realized assortment in weeks 2–4 differs from the first week after reopening. We test the joint null

$$H_0 : \beta_2 = \beta_3 = \beta_4 = 0,$$

which corresponds to no systematic change in the assortment across the four post-reopening weeks.

The estimates show no statistically detectable change in core-product coverage, but they reject a constant path for standardized product breadth and pre-menu overlap. Product breadth is 1.66 products lower by week 4, and pre-menu overlap is 11.11 percentage points lower. Assortment stability alone therefore cannot rule out differential return timing.

**Return timing and customer-specific novel opportunity.** We next test the channel directly among treated customers. A returner is a treated customer whose first purchase at the reopened focal store occurs within 28 days of reopening. Of 5,349 low-intention and 2,041 high-intention treated customers, 1,134 (21.2%) and 1,268 (62.1%), respectively, return within this window. High-intention returners return 2.87 days earlier after conditioning on closure-event fixed effects ( $SE = 0.29$ ;  $p < 0.001$ ). Their median return is day 7 rather than day 11, and 53.6% return in week 1 compared with 36.9% of low-intention returners.

Table W10: Evolution of realized assortment after reopening

|                                 | Core-product coverage<br>(1) | Products per 50 baskets<br>(2) | Pre-menu overlap<br>(3) |
|---------------------------------|------------------------------|--------------------------------|-------------------------|
| Week 2 relative to week 1       | 0.0059<br>(0.0185)           | 0.3019<br>(0.7351)             | −0.0100<br>(0.0129)     |
| Week 3 relative to week 1       | −0.0648<br>(0.0544)          | −0.7031<br>(0.6006)            | −0.0763<br>(0.0410)     |
| Week 4 relative to week 1       | −0.0840<br>(0.0537)          | −1.6600<br>(0.5552)            | −0.1111<br>(0.0403)     |
| Joint equality test, $p$ -value | 0.3821                       | 0.0106                         | 0.0175                  |
| Week 1 mean                     | 0.7687                       | 29.2581                        | 0.6120                  |
| Observations                    | 72                           | 64                             | 72                      |
| Treated stores                  | 18                           | 16                             | 18                      |

Notes: The sample contains the treated stores during the first four fixed seven-day periods after reopening. Core-product coverage is the share of products in the store’s core pre-closure set that sell in the current week. Products per 50 baskets averages the number of distinct products across 100 samples of 50 customer-day baskets drawn without replacement. Pre-menu overlap is the Jaccard similarity between products sold in the current week and the store’s pre-closure product set. Week 1 is the omitted category in equation (13). Standard errors in parentheses use CRV1 clustering by closure event. The joint row reports restricted Rademacher wild-cluster  $p$ -values based on 9,999 replications for the null that the coefficients for weeks 2, 3, and 4 are jointly zero. Columns (1) and (3) contain all 18 treated stores; column (2) requires at least 50 baskets in every week and therefore contains 16.

For each returner, we construct the realized product set in each of the four candidate return weeks. To prevent the customer’s observed purchase from mechanically entering the opportunity set, we remove any product sold only to that customer in the store-week; products also purchased by another customer remain. We define a product as personally novel if the customer had not purchased it anywhere in the chain by the reopening date. Let  $M_{iew}$  denote either the leave-one-customer-out novel-product share or count for member  $i$ , event  $e$ , and candidate return week  $w$ . Our timing measure is

$$D_{ie} = M_{ie,w(i)} - \frac{1}{4} \sum_{w=1}^4 M_{iew}, \quad (14)$$

where  $w(i)$  is the customer’s actual return week. Subtracting the same customer’s four-week mean removes persistent high–low differences in product history and retains only the opportunity change associated with returning in one week rather than another. We regress  $D_{ie}$  on the high-intention indicator and closure-event fixed effects, clustering by closure event.

Table W11: Return timing and personally novel product opportunity

|                          | Days to first return             | Novel-product share             | Novel-product count             |
|--------------------------|----------------------------------|---------------------------------|---------------------------------|
|                          | (1)                              | (2)                             | (3)                             |
| High minus low intention | −2.8724<br>(0.2936)<br>[< 0.001] | −0.0010<br>(0.0006)<br>[0.0930] | −0.2756<br>(0.3848)<br>[0.4836] |
| Low-intention mean       | 12.1570                          | −0.0005                         | 1.5780                          |
| High-intention mean      | 9.0331                           | −0.0016                         | 1.0562                          |
| Observations             | 2,402                            | 2,325                           | 2,402                           |
| Closure-event clusters   | 18                               | 17                              | 18                              |

Notes: The sample contains treated customers whose first focal-store purchase occurs within 28 days after reopening. Column (1) reports days from reopening to that purchase. Columns (2) and (3) report the timing deviations in equation (14); means in those columns are therefore deviations from each customer’s four-week average opportunity, not levels. The share is the fraction of the leave-one-customer-out realized product set that the customer had never purchased by reopening, and the count is the number of such products. The share analysis requires a nonempty leave-one-customer-out product set in all four weeks. All specifications include closure-event fixed effects. Standard errors clustered by closure event are in parentheses, and two-sided  $p$ -values are in brackets.

The high-minus-low timing deviation is −0.10 percentage points for the novel-product share (95% CI: −0.22 to +0.02) and −0.28 products for the count (95% CI: −1.09 to +0.54). The share estimate has the direction required by the alternative explanation but is only about one-fortieth of the −4.15-percentage-point main novelty DDD. The count estimate is statistically indistinguishable from zero and small relative to roughly 67 personally novel products in the realized return-week sets. Differential return timing therefore produces little differential exposure to personally novel products in the observed transaction environment, weighing against it as a quantitatively important explanation for the main result.

**Comparison with matched control stores.** As an additional check, we examine whether treated stores experience an unusual change in realized assortment relative to their matched control stores. This analysis uses the 18 treated stores and their five matched control stores and constructs four fixed seven-day periods before closure and four fixed seven-day periods after reopening. For assortment outcome  $A_{est}$  at store  $s$ , event  $e$ , and relative week  $t$ , we estimate

$$A_{est} = \beta(\text{TreatedStore}_{es} \times \text{Post}_{et}) + \alpha_{es} + \lambda_{et} + \epsilon_{est}, \quad (15)$$

where  $\alpha_{es}$  denotes event-store fixed effects and  $\lambda_{et}$  denotes event-by-relative-week fixed effects. The specification compares the change in assortment at the treated store with the contemporaneous change among its matched controls.

The matched-store estimates provide no evidence of a relative contraction after reopening. Core-product coverage falls by 1.29 percentage points more at treated stores than at controls (SE = 3.41 percentage points;  $p = 0.710$ ). Standardized product breadth increases by 0.31 products per 50 baskets (SE = 0.52;  $p = 0.562$ ), while pre-menu overlap increases by 1.46 percentage points (SE = 2.50 percentage points;  $p = 0.567$ ). The corresponding event-study estimates do not show a common relative post-reopening decline, and the joint pretrend tests do not reject for any of the three measures.

Table W12: Realized assortment at reopening stores and matched controls

|                                             | Core-product coverage<br>(1)    | Products per 50 baskets<br>(2) | Pre-menu overlap<br>(3)        |
|---------------------------------------------|---------------------------------|--------------------------------|--------------------------------|
| Post $\times$ Treated store                 | −0.0129<br>(0.0341)<br>[0.7097] | 0.3063<br>(0.5178)<br>[0.5619] | 0.0146<br>(0.0250)<br>[0.5665] |
| Immediate reopening, relative to week<br>−1 | 0.0205<br>(0.0365)<br>[0.5806]  | 0.1935<br>(0.8753)<br>[0.8277] | 0.0495<br>(0.0293)<br>[0.1097] |
| Joint pretrend $p$ -value                   | 0.1103                          | 0.4123                         | 0.5218                         |
| Control pre-period mean                     | 0.9240                          | 28.2045                        | 0.7513                         |
| Observations                                | 736                             | 723                            | 736                            |
| Event-store pairs                           | 92                              | 92                             | 92                             |
| Closure-event clusters                      | 18                              | 18                             | 18                             |

Notes: Core-product coverage is the share of products in a store's core pre-closure set that sell in the current week. Products per 50 baskets averages the number of distinct products across 100 samples of 50 customer-day baskets drawn without replacement.

Pre-menu overlap is the Jaccard similarity between products sold in the current week and the store's pre-closure product set. Standard errors clustered by the 18 closure events are in parentheses, and two-sided  $p$ -values are in brackets. Product sales measure realized assortment and need not equal the product set displayed or available to every consumer.

The matched-store analysis addresses a different question from the return-timing test. It asks whether closure and reopening are associated with an unusual store-level assortment change relative to contemporaneous controls. The four-week path shows that realized assortment subsequently evolves, so the direct diagnostic for the alternative is instead the customer-specific timing deviation in equation (14). The transaction data cannot reveal the exact product set presented to each consumer, the return analysis is descriptive among customers who return within 28 days, and no structural model maps the opportunity-share contrast into the purchase-novelty outcome. Subject to these limitations, the small observed timing-induced differ-



ences weigh against differential return timing as a quantitatively important explanation rather than ruling it out categorically.

## D Additional Results

### D.1 Complete Main Specifications

The main text reports the group-specific effects and DDDs that carry the economic interpretation. This section reports the full coefficient vectors from the underlying regressions. Table W13 presents equation (9) for the two main outcomes.

Table W13: Complete binary predicted-intention specifications

|                                                                        | Purchase frequency<br>(1)          | Novelty-seeking<br>(2)              |
|------------------------------------------------------------------------|------------------------------------|-------------------------------------|
| Post $\times$ Treated ( $\delta^B$ )                                   | .0009<br>(.0024)<br>[.7107]        | .0313<br>(.0178)<br>[.0976]         |
| Post $\times$ High predicted intention ( $\beta$ )                     | -.0344<br>(.0030)<br>[<.001]       | .0381<br>(.0081)<br>[<.001]         |
| Post $\times$ Treated $\times$ High predicted intention ( $\delta^D$ ) | <b>.0058</b><br>(.0032)<br>[.0860] | <b>-.0415</b><br>(.0184)<br>[.0380] |
| Pre-period outcome mean                                                | .0493                              | .4978                               |
| Observations                                                           | 321,184                            | 99,644                              |
| $R^2$                                                                  | .562                               | .416                                |
| Within $R^2$                                                           | .013                               | .001                                |
| Member-closure fixed effects                                           | Yes                                | Yes                                 |
| Relative-period fixed effects                                          | Yes                                | Yes                                 |
| Calendar-month fixed effects                                           | Yes                                | Yes                                 |

Notes: This table reports every coefficient from equation (9). Post  $\times$  Treated is the low-predicted-intention treated-control post contrast. Post  $\times$  High captures the high-minus-low post shift common to treated and control customers and is not a treatment effect. The triple interaction is the high-minus-low DDD and is shown in boldface. The closure window is omitted. Novelty-seeking is conditional on a purchase in the window. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. Table 3 reports the directly estimated high-group effects and gives detailed outcome definitions.

The common post shift,  $\beta$ , is large for both outcomes. high-predicted-intention customers have lower post-period purchase frequency and higher post-period novelty-seeking than low-predicted-intention customers even after pooling treated and control observations. These coefficients reflect the relation between predicted purchase timing and post-period behavior; they are not closure effects. The DDD removes this common high-minus-low movement and isolates the additional treated high-group response under the identifying assumptions. The algebraically equivalent group-effect parameterization in Table 3 then combines  $\delta^B$  and  $\delta^D$  with covariance-aware standard errors.

### D.1.1 Continuous predicted-intention score

To avoid relying on the 0.5 classification threshold, we replace  $\text{High}_{ie}$  with the mean-centered predicted purchase-intention score  $\tilde{s}_{ie}$ :

$$Y_{iet} = \delta_0^B \cdot \text{Post} \times \text{Treated} + \beta^S \cdot \text{Post} \times \tilde{s}_{ie} + \delta^S \cdot \text{Post} \times \text{Treated} \times \tilde{s}_{ie} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}. \quad (16)$$

Because the score is centered,  $\delta_0^B$  is the treated-control post contrast at the mean predicted intention. The coefficient  $\beta^S$  is the common post-score gradient, while  $\delta^S$  is the change in the treated-control post contrast associated with a one-unit increase in predicted purchase intention.

Table W14: Complete continuous predicted-intention specifications

|                                                                   | Purchase frequency<br>(1)          | Novelty-seeking<br>(2)              |
|-------------------------------------------------------------------|------------------------------------|-------------------------------------|
| Post $\times$ Treated                                             | .0023<br>(.0028)<br>[.4337]        | .0190<br>(.0151)<br>[.2265]         |
| Post $\times$ Centered predicted-intention score                  | -.0568<br>(.0046)<br>[<.001]       | .0533<br>(.0149)<br>[.0024]         |
| Post $\times$ Treated $\times$ Centered predicted-intention score | <b>.0076</b><br>(.0051)<br>[.1522] | <b>-.0452</b><br>(.0276)<br>[.1199] |
| Observations                                                      | 321,184                            | 99,644                              |
| $R^2$                                                             | .565                               | .416                                |
| Within $R^2$                                                      | .018                               | .000                                |
| Member-closure fixed effects                                      | Yes                                | Yes                                 |
| Relative-period fixed effects                                     | Yes                                | Yes                                 |
| Calendar-month fixed effects                                      | Yes                                | Yes                                 |

Notes: The predicted-intention score is centered at its estimation-sample mean. Post  $\times$  Treated therefore gives the treated-control post contrast at the mean score. The score coefficient is the post-period score gradient common to treated and control customers. The triple interaction, shown in boldface, is the focal coefficient and shows how the treated-control post contrast changes with the predicted-intention score. All remaining conventions and outcome definitions follow Table W13. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets.

The continuous estimates preserve the binary results' sign pattern. Higher predicted purchase intention is associated with a larger treated-control recovery in purchase frequency and a smaller treated-control post response in novelty-seeking. For a 10-percentage-point increase in the predicted-intention score, the fitted DDD changes by +0.08 percentage points per day for purchase frequency and  $-0.45$  points for novelty-seeking. Neither slope interaction is statistically distinguishable from zero at the 5% level under closure-

clustered inference. The purchase-frequency pretrend caveat remains, while the novelty-seeking score-pretrend test does not reject equality.

## D.2 Alternative Outcome Definition: New-Product-Seeking

The main novelty-seeking outcome defines exploration relative to the customer’s own purchase history. As a robustness check, we instead ask whether customers select products that have appeared recently in the transaction data. Let  $F_j$  be product  $j$ ’s first observed sale date and let  $\mathcal{W}_{et}$  and  $\mathcal{W}_{e,t-1}$  be the current and immediately previous event windows. We define new-product-seeking as

$$Y_{iet}^{NP} = \frac{|\{j \in J_{iet} : F_j \in \mathcal{W}_{e,t-1} \cup \mathcal{W}_{et}\}|}{|J_{iet}|}. \quad (17)$$

This measure is the share of distinct products purchased in the current window whose first observed sale falls in the current or immediately previous event window. It proxies for recently introduced products, but first observed sale need not be the formal catalog launch date. Like novelty-seeking, it is defined only in windows with a purchase.

The binary specification yields a 0.80-point low-intention effect and a 3.37-point decline for high-predicted-intention customers. Their difference is  $-4.17$  percentage points (SE 0.89;  $p < 0.001$ ), or 28.6% of the pre-period mean. The continuous-score specification leads to the same conclusion: a 10-percentage-point increase in predicted purchase intention lowers the treated-control post contrast by 0.49 percentage points ( $p = 0.006$ ). The high-minus-low pretrend tests do not reject for either specification ( $p = 0.114$  for the binary measure and  $p = 0.377$  for the continuous score). The robustness check therefore supports the main novelty-seeking result using a definition tied to recently appearing products rather than the customer’s own history.

## D.3 Novelty-Seeking among Non-Coffee Products

This exercise examines whether the main product-choice pattern is confined to choices among coffee products. We classify products using the commodity-description fields in the transaction data. Each product line belongs to one mutually exclusive source category. The broad non-coffee definition combines non-coffee beverages, food and other non-coffee items; the consumables definition retains beverages and food; and the beverage-only definition retains non-coffee drinks. We identify products jointly by source category

Table W15: Robustness to using new-product-seeking

|                                                                     | Binary high-intention<br>(1)        | Continuous intention score<br>(2)   |
|---------------------------------------------------------------------|-------------------------------------|-------------------------------------|
| <i>Panel A. Group effects and incremental interruption contrast</i> |                                     |                                     |
| Low predicted purchase intention                                    | .0080<br>(.0113)<br>[.4871]         | —                                   |
| High predicted purchase intention                                   | -.0337<br>(.0145)<br>[.0331]        | —                                   |
| High minus low DDD                                                  | <b>-.0417</b><br>(.0089)<br>[<.001] | —                                   |
| <i>Panel B. Complete regression coefficients</i>                    |                                     |                                     |
| Post × Treated                                                      | .0080<br>(.0113)<br>[.4871]         | -.0033<br>(.0120)<br>[.7851]        |
| Post × Predicted intention                                          | .0300<br>(.0045)<br>[<.001]         | .0468<br>(.0099)<br>[<.001]         |
| Post × Treated × Predicted intention                                | <b>-.0417</b><br>(.0089)<br>[<.001] | <b>-.0486</b><br>(.0156)<br>[.0063] |
| Pre-period outcome mean                                             | .1456                               | .1456                               |
| Observations                                                        | 99,644                              | 99,644                              |
| $R^2$                                                               | .393                                | .393                                |
| Within $R^2$                                                        | .0008                               | .0007                               |
| Member-closure fixed effects                                        | Yes                                 | Yes                                 |
| Relative-period fixed effects                                       | Yes                                 | Yes                                 |
| Calendar-month fixed effects                                        | Yes                                 | Yes                                 |

Notes: Column (1) uses the binary high-predicted-intention indicator; column (2) uses the mean-centered predicted-intention score. In Panel A, the low- and high-intention rows are directly estimated treated-control post contrasts, and the high-minus-low row is their DDD. These discrete group effects are not defined for column (2). In Panel B, Post × Predicted intention is the common post shift across treated and control customers; it is not a treatment effect. Boldface marks the focal DDD coefficients. In column (2), a one-unit change corresponds to moving across the full predicted-intention scale. Standard errors, clustered by closure event (18 clusters), are in parentheses; p-values are in brackets. All regressions omit the closure window.

and exact description so that identical descriptions appearing in different categories remain distinct. Within each scope, novelty-seeking is the share of distinct qualifying products that the customer had not purchased previously and is observed only in windows containing a qualifying purchase.

Table W16 reports the collapsed DDD estimates and the corresponding sample-entry diagnostics.

The collapsed DDDs are negative across all three product scopes but imprecisely estimated. The all-non-coffee estimate is −4.13 percentage points, with a 95% confidence interval from −11.31 to 3.05 points. Removing non-consumable items has essentially no effect because these products account for few qualifying

Table W16: Novelty-Seeking among Non-Coffee Products

|                                                         | All non-coffee  | Non-coffee consumables | Non-coffee beverages |
|---------------------------------------------------------|-----------------|------------------------|----------------------|
| <i>Panel A. Novelty-seeking conditional on purchase</i> |                 |                        |                      |
| High-minus-low DDD                                      | −4.13<br>(3.40) | −4.11<br>(3.39)        | −6.26<br>(4.12)      |
| CRV1 <i>p</i> -value                                    | .242            | .242                   | .147                 |
| Restricted wild-cluster <i>p</i> -value                 | .225            | .230                   | .119                 |
| Joint pretrend <i>p</i> -value                          | .006            | .006                   | .226                 |
| Observations                                            | 30,444          | 30,418                 | 21,433               |
| <i>Panel B. Probability of any qualifying purchase</i>  |                 |                        |                      |
| High-minus-low DDD                                      | −1.70<br>(1.00) | −1.75<br>(1.01)        | −2.24<br>(0.90)      |
| CRV1 <i>p</i> -value                                    | .107            | .102                   | .023                 |
| Joint pretrend <i>p</i> -value                          | .093            | .088                   | .121                 |
| Observations                                            | 321,184         | 321,184                | 321,184              |

Notes: Estimates and standard errors are in percentage points. All non-coffee combines non-coffee beverages, food and other non-coffee items. Non-coffee consumables combines beverages and food. Panel A defines novelty-seeking as the share of distinct products in the selected scope that the customer had not purchased before the window; it is observed only in windows containing a qualifying purchase. Panel B uses the full member-closure-period panel and indicates whether the window contains such a purchase. All regressions use the same binary DDD specification, fixed effects and 18 closure-event clusters as the main analysis.

The joint pretrend row tests whether the DDD coefficients in relative periods −4, −3 and −2 are jointly zero, with period −1 omitted.

purchasing windows. The beverage-only estimate is more negative, but its confidence interval also includes zero.

The dynamic and sample-entry diagnostics prevent a causal interpretation of these estimates. For the all-non-coffee and consumables outcomes, the joint lead tests reject parallel triple trends. The rejection is driven primarily by relative period −3, in which the event-study DDD is +10.33 percentage points ( $p = 0.005$ ); the post-reopening coefficients also alternate in sign. The beverage-only novelty specification does not reject its pretrend, but its entry DDD shows that high-predicted-intention treated customers become differentially less likely to contain a qualifying purchase. Because novelty-seeking is conditional on such a purchase, this change may alter the composition of the observed sample. The exercise therefore provides a product-scope sensitivity check rather than independent confirmation of the main causal result.

#### D.4 Accounting for Local Competition

The store-address data contain the number of locations classified as cafés within 500 and 1,500 meters of each of the chain's 260 stores. We use the 500-meter count as the primary measure and the 1,500-meter count as a radius sensitivity. Both fields match all 108 stores in the closure-event design. We assign the counts

through each member-event’s preferred pre-closure store, reconstructed using the same five-purchase-day and 80% purchase-share requirements as in the main analysis. The source does not identify café brands, operating status, its underlying point-of-interest provider or the date at which the counts were measured. The measures are therefore fixed proxies for local café density in the regressions; they do not directly observe other-brand competition or competitor closures during the treatment period.

Treated stores have an average of 6.00 cafés within 500 meters, compared with 12.54 among control stores, for a treated-minus-control standardized mean difference of  $-0.397$ . Their observed ranges overlap from 0 to 19 cafés. This interval contains all 18 treated stores and 75 of the 90 distinct control stores. To allow this fixed store characteristic to change the post-reopening comparison, let  $\tilde{C}_s$  denote the standardized café count assigned through preferred store  $s$ . We add the following hierarchy to equation (9):

$$\text{Post}_{et} \left[ \kappa_C \tilde{C}_s + \kappa_{TC} (\text{Treated}_{ie} \tilde{C}_s) + \kappa_{HC} (\text{High}_{ie} \tilde{C}_s) + \kappa_{THC} (\text{Treated}_{ie} \text{High}_{ie} \tilde{C}_s) \right]. \quad (18)$$

The original Post-by-Treated, Post-by-High and triple-interaction terms remain in the regression. Because density is standardized across unique member-events, the coefficient on  $\text{Post} \times \text{Treated} \times \text{High}$  is the fitted DDD at mean density. At standardized density  $c$ , the fitted DDD is  $\delta^D + \kappa_{THC}c$ , and its standard error uses the covariance between the two coefficients.

The continuous specifications preserve the negative DDD at their member-event means under both radii and transformations. In the primary 500-meter specification, the density gradient is imprecisely estimated, and the fitted DDD remains negative at each of the three prespecified points. Removing one closure and its five matched controls at a time produces fitted DDDs ranging from  $-7.10$  to  $-3.18$  points at a count of 2 and from  $-3.25$  to  $-1.15$  points at a count of 13. The sign is therefore not attributable to any single closure. Using new-product-seeking rather than member-first novelty also preserves the pattern: the fitted DDD is  $-3.85$  points at mean 500-meter density and ranges from  $-3.10$  to  $-4.12$  points across counts 2, 5 and 13.

Two additional checks address local support and unobserved additive shocks. A descriptive restriction to stores with no more than three nearby cafés retains 42 design stores, including 9 treated stores, and yields a DDD of  $-5.39$  points. This comparison has 16 contributing closure clusters, and only 8 matched sets contain qualifying stores on both treatment sides, so its confidence interval is necessarily wide. Preferred-store-by-event-by-post fixed effects absorb arbitrary additive post-reopening shocks shared by members assigned to the same store. The resulting DDD is  $-4.45$  points. These fixed effects do not absorb a local shock that

Table W17: Novelty-seeking DDDs across measures of local café density

| Specification or density point                                 | Estimate | SE   | 95% CI         | Wild-cluster $p$ -value |
|----------------------------------------------------------------|----------|------|----------------|-------------------------|
| <i>Panel A. Collapsed specifications</i>                       |          |      |                |                         |
| Paper-facing baseline                                          | −4.15    | 1.84 | [−8.04, −0.26] | .045                    |
| Raw 500m count, member-event mean                              | −3.33    | 1.43 | [−6.35, −0.31] | .090                    |
| $\log(1 + 500\text{m count})$ , member-event mean              | −3.92    | 1.76 | [−7.64, −0.20] | .065                    |
| Raw 1,500m count, member-event mean                            | −4.04    | 1.64 | [−7.51, −0.57] | .038                    |
| $\log(1 + 1,500\text{m count})$ , member-event mean            | −4.11    | 1.59 | [−7.46, −0.76] | .029                    |
| 500m count $\leq 3$ (descriptive)                              | −5.39    | 3.65 | [−13.17, 2.40] | .178                    |
| Preferred-store $\times$ event $\times$ post fixed effects     | −4.45    | 1.71 | [−8.07, −0.84] | .016                    |
| <i>Panel B. Fitted DDDs in the raw 500-meter specification</i> |          |      |                |                         |
| 500m count = 2                                                 | −5.03    | 3.15 | [−11.67, 1.61] | .161                    |
| 500m count = 5                                                 | −4.39    | 2.42 | [−9.50, 0.71]  | .109                    |
| 500m count = 13                                                | −2.70    | 1.25 | [−5.35, −0.06] | .213                    |
| Change in DDD per SD of 500m density                           | 3.00     | 3.86 | [−5.15, 11.14] | .490                    |

Notes: Estimates and standard errors are in percentage points. Density-interacted estimates are evaluated using the complete Post-interaction hierarchy in equation (18). Counts 2, 5 and 13 are the 25th percentile, median and 75th percentile among the 108 design stores. All full-sample regressions contain 99,644 observations and 18 closure-event clusters. The low-density row contains 41,175 observations and 16 clusters. The preferred-store fixed-effect row adds preferred-store-by-event-by-post fixed effects, which absorb additive local shocks common to high- and low-intention members assigned to the same store. Standard errors and confidence intervals use CRV1 clustering by closure event. Restricted wild-cluster  $p$ -values use 9,999 Rademacher draws.

changes high- and low-predicted-intention behavior differently within the same store.

Finally, we interact the full density hierarchy with relative-period indicators, using period  $-1$  as the reference. The estimated density gradients in periods  $-4$ ,  $-3$  and  $-2$  are 0.1309, 0.0906 and 0.0695, and their joint closure-clustered test rejects equality to zero ( $p = 0.002$ ). This differential pre-closure movement limits a causal interpretation of the density-interacted specification. By comparison, an event study with preferred-store-by-event-by-period fixed effects does not reject its three DDD leads jointly ( $p = 0.880$ ). Taken together, the exercise shows that the negative collapsed DDD is not confined to high observed café density, while the dynamic diagnostic prevents the proxy from isolating a distinct competition channel.